The Compiler Closure of TFPT

The discrete compiler, its status, and how to read the document set —
a status assessment and reading guide

June 12, 2026

In one sentence

The theory tips from "many sectors with surprising hits" to "*one small discrete compiler generates those sectors*": E_8 is *built* from two TFPT atoms and sits at the centre as the audit hull; the flavor matrix, the fine-structure constant and the scale grammar follow as *fixed points* from exactly two inputs $c_3 = \frac{1}{8\pi}$ and $g_{\text{car}} = 5$.

The TFPT document set — what is treated where

Plain language: TFPT is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard Model, the constants and the scale grammar. The development is **five short documents plus Appendix H**, best read in order (this is the entry point; a detailed map is in §1):

1. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and proof ledger.
2. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction.
3. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport, the worked closures, and gravity/QG as the seam response.
4. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.
5. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.

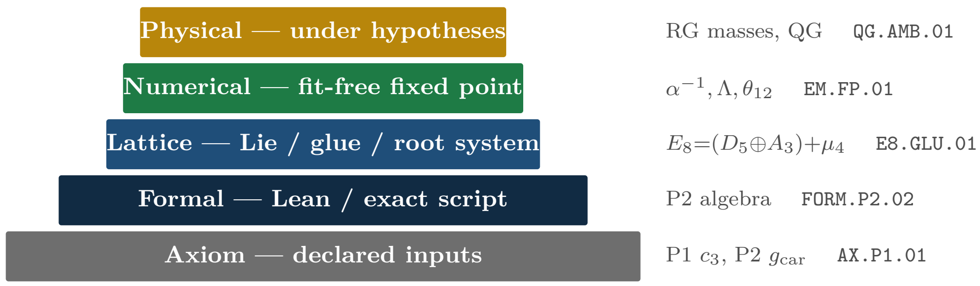
H. `tfpt_horizon_readouts` — **Appendix H** (unit-system reframe, *not* new physics): one seam constant as the universal horizon thermal code.

O. `origin_theory` — **Origin Theory**: why no free fundamental number remains. The $(g_{\text{car}}, N_{\text{fam}})$=$(5, 3)$ skeleton, the triply-forced 8 (geometry = lattice = gravity), the order-30 Coxeter cycle, one boundary transport for flavor and horizon, and the gapped *unique attractor*. [I]/[L] core (`v53`–`v56`) plus one honestly-typed [P] cyclic interpretation.

You are reading document #1.

Contents

Reading the markers (refined). The former overloaded "[M]" is split into the grade it actually carries, so a reviewer can see at a glance which claims are exact, which are formal, and which are conditional: [I] *exact identity* (e.g. $240 = 16 \cdot 5 \cdot 3$); [L] *Lie/lattice theorem* (e.g. $D_5 \oplus A_3 + \mu_4 = E_8$); [F] *formalised* (Lean- or script-verified); [N] *numerical fixed point* (reproduced, stable); [P] *physical/conditional* (follows under named hypotheses); [A] *axiom or open*.



The claim stack (bottom = most certain/fewest, top = conditional). Each claim sits on exactly one level; the bottom two are inputs, the middle two are proved/reproduced, only the top is conditional. The single source of truth for which claim sits where is `verification/status_ledger.csv` (Fig. 1).

No-free-pattern rule. To keep the structure from drowning in decorative coincidences, an identity enters the *main text* only if it (1) is a necessary step in a derivation, (2) kills an alternative, (3) is ablation-relevant, (4) links two previously independent modules, or (5) is experimentally tested. Everything else is explicitly demoted to an *audit fingerprint* [I] (clearly boxed as such) — so the load-bearing beams stay visible and the ornament never gets promoted to spine.

Admissible-invariant classes (the harder filter). Sharper still: a *load-bearing* integer may only come from one of seven typed sources — (i) a symmetric function $e_k(a)$ of the anchor $a = (1, 1, 2)$; (ii) a power sum $p_n(a)$ or difference $p_{n+1} - p_n$; (iii) a determinant, spectrum, Smith normal form or trace of (R, Q, K, L); (iv) an anchor quadratic form $u^\top M v$ with $u, v \in \{\mathbf{1}, a\}$; (v) a Plücker coordinate of the anchor plane $\langle \mathbf{1}, a \rangle$; (vi) an E_8 Casimir degree or branching block; (vii) a pre-registered experimental observable. Anything else is an audit fingerprint or a frontier item, never spine. This is what converts TFPT from *pattern hunting* ("look, another number") into a typed compiler: *fewer admissible kinds of explanation*, not more matches.

Reviewer path (read in this order; the rest is support)

1. the architecture: the two axioms $\{c_3, g_{\text{car}}\}$ and the two-engine picture (§1, the figures);
2. the E_8 glue $E_8 = (D_5 \oplus A_3) + \mu_4$ — document 1, Part II, machine-checked (`verification/v1_e8_glue.py`);
3. the α^{-1} fixed point plus ablation (`v3_em_alpha.py`);
4. the **single status ledger** `verification/status_ledger.csv` — the source of truth for

every claim's grade, location, dependency and script;

5. the honest frontier list (document 4).

Everything carries a claim ID (e.g. `E8.GLU.01`, `EM.FP.01`) that resolves to a row of the ledger. **If the text and the ledger ever disagree, the ledger wins.**

1 What this is about: the jump in construction principle

In plain words — what we found, and why it matters

What we found. TFPT reproduces dozens of real numbers of physics (particle masses, the fine-structure constant, dark-energy and inflation values) from a small set of inputs — and the compiler closure says *why* the same handful of small integers $(2, 3, 5, 16, \dots)$ reappears in every sector: those numbers are not separate lucky hits. A single tiny "machine" — fed by just **two inputs** (a boundary number $\frac{1}{8\pi}$ and a five-slot carrier) — builds the exceptional symmetry E_8, and E_8 is the unimodular "audit hull" that classifies the discrete Standard-Model readout: the constants, the flavor skeleton and the gravity/cosmology regime are then read off by projection and boundary dressing (not "printed" for free — where a scheme or analytic interface is needed, it is named).

Why this is simple. There are not many separate derivations that happen to share numbers — there is *one generator and a handful of corollaries*, and the number of independent structural assumptions is **two**.

How plausible it is. A theory becomes more believable when it *explains more from less*. Here the two inputs are even *over-determined*: the machine reproduces the very two numbers it started from (the dashed bootstrap loop in the architecture diagram, §2), so there is *no free dial left to tune*. The only genuinely free, continuous ingredient that remains is the number π.

Sharper still: one anchor $a = (1, 1, 2)$, and E_8 as a *compiler hull* [I]/

The two axioms are not even independent: they are the *elementary symmetric polynomials* of the single parabolic anchor $a = (1, 1, 2)$ (the exponents-at-infinity / splitting type $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$):

$$e_1(a) = 4 = |\mu_4|, \qquad e_2(a) = 5 = g_{\text{car}}, \qquad e_3(a) = 2 = |\mathbb{Z}_2|, \qquad c_3 = \frac{1}{2e_1(a)\pi} = \frac{1}{8\pi}.$$

Its *power sums* $p_n(a) = 1^n+1^n+2^n = 2+2^n$ generate the big Lie data directly: $p_1{=}4{=}|\mu_4|$, $p_2{=}6{=}|R^+(A_3)|$, $p_3{=}10{=}\mathcal{A}_\Lambda$, and

$$|R(E_8)| = p_1p_2p_3 = 240, \quad \dim E_8 = p_1p_2p_3 + (p_4 - p_3) = 248, \quad p_4 - p_3 = 8 = \operatorname{rank} E_8,$$

with the binary ladder $p_{n+1} - p_n = 2^n$. So the inputs collapse from $\{c_3, g_{\text{car}}\}$ to $\boxed{a = (1, 1, 2) \text{ plus } \pi}$. [`verification/v23_anchor_generator.py`]

And a is itself half-forced ([L]/[P]): on $\mathbb{P}^1$ every bundle splits (Birkhoff–Grothendieck), so once $|\mu_4| = 4$, $N_{\text{fam}} = 3$, $\deg E = -4$ are set, the three positive anti-degrees are positive integers summing to 4, and $4 = 2 + 1 + 1$ is the *unique* partition into three positive parts. The genuine remaining axiom is therefore not "why $a = (1, 1, 2)$" but "why the carrier $g_{\text{car}} = 5$ / $(|\mu_4|, N_{\text{fam}}) = (4, 3)$" — that interface (with c_3) is the [A] input layer.

What E_8 is (and is not). In TFPT E_8 is *not* an unbroken physical gauge group: it is the unimodular *audit / compiler hull* that classifies the admissible discrete charge and residue structures. The Standard Model is a *readout* after projection, not a direct E_8 gauge theory; the Lorentz signature appears only after projection; fermions are not "everything in the adjoint". This is exactly why the known no-go results against literal E_8 world-formulas (Distler–Garibaldi;

Coleman–Mandula) do *not* bite: they constrain E_8 as a spacetime gauge symmetry, which TFPT does not claim. The defensible claim is: *TFPT is the unique parabolic compilation of the anchor* $a = (1, 1, 2)$ *into the* E_8 *audit hull.*

In one line:

The anchor generates the syntax; the boundary generates the scale; E_8 checks the consistency.

i.e. the discrete content (carrier $D_5 \oplus A_3 + \mu_4$, the operators R, Q, K, L) flows from $a = (1, 1, 2)$; the continuous content ($c_3 = \frac{1}{8\pi}$, φ_0, α^{-1}, the exponential scales) flows from π; and E_8 is the unimodular *checksum container*, not the world group.

The anchor ratio triad ([verification/v119_review_validation_2.py]). The three elementary symmetric values, normalised by the family count, are the theory's three recurring fractions [I]:

$$\frac{e_3}{p_0} = \frac{2}{3} \text{ (Koide branch / sheet ratio)}, \qquad \frac{e_1}{p_0} = \frac{4}{3} \text{ (seed gain)}, \qquad \frac{e_2}{p_0} = \frac{5}{3} \text{ (carrier branch)}$$

— the canonical flow's critical points are exactly $(2, 5) = (e_3, e_2)$ with inflection $\frac{7}{2} = \frac{e_3+e_2}{2}$: the fractions $\frac{2}{3}, \frac{4}{3}, \frac{5}{3}$ are not scattered numbers but the three normalised anchor coefficients. Further micro-identities [I]: $\Omega_{\text{adm}} = 2p_1p_2 = 48$, $10b_1 = 1 + p_1p_3 = 41$, $\Delta_Y = e_2^2 = p_0^2 + 2\,\text{rank}\,E_8 = 25$, $h(E_8) = e_3p_0e_2 = 30$, $\text{rank}\,E_8 = p_0 + e_2$.

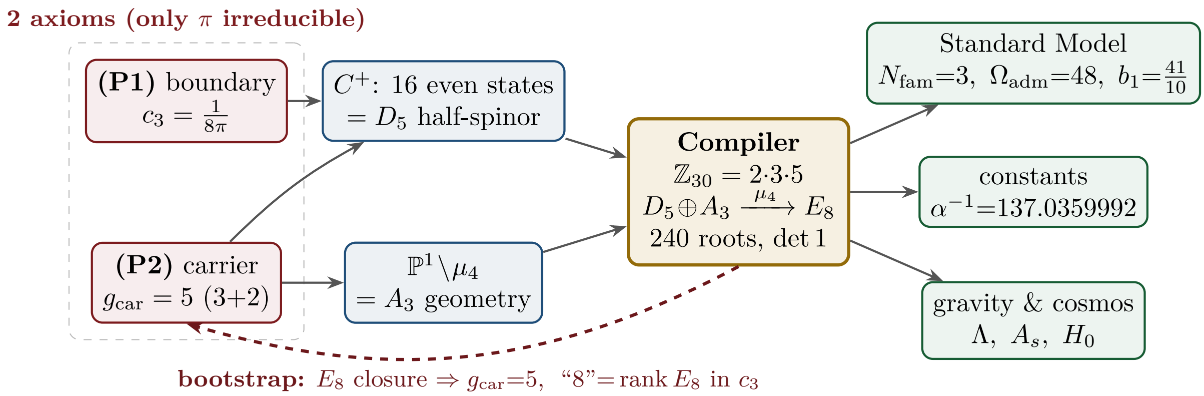
That is now the case. The whole development is consolidated into **five companion documents** (plus this introduction), organised in layers that build on one another; this introduction is the entry point and reading guide. Each document collects the earlier notes of its layer into self-contained parts:

Document	What it contains (former notes now as parts)
`tfpt_1_architecture_e8`	**Core.** The two axioms $\{c_3, g_{\text{car}}\}$, the derivation map and the EM fixed point (Part I); the Coxeter–cyclotomic compiler $\mathbb{Z}_{30}$=2·3·5, the explicit $D_5 \oplus A_3 + \mu_4$ E_8 construction, the machine-checkable E_8 appendix and the group-level glue $(\text{Spin}(10) \times SU(4))/\Delta\mathbb{Z}_4$ (Part II).
`tfpt_2_standard_model`	**Standard Model.** The full SM in one φ_0^{ret}-ladder formula (Part I); the flavor block from parabolic weights and the transport theorem, H2 as a spectral selector (Part II); the worked closures — explicit gap, APS–Fredholm, truncation/n_s, r, Starobinsky M, derived θ_{12}, quark c, the H2 splitting type, Λ-typing (Part III).
`tfpt_3_e8_audit_bootstrap`	E_8 **audit & bootstrap.** The seven E_8 slices as an audit raster ($E_6 \times A_2$ reads the flavor matrix) (Part I); the old cascade D=60−2n as the same even-integer spine (Part II); the Möbius bootstrap — g_{car}=5 and the "8" in c_3 as overdetermined fixed points, only π irreducible (Part III).
`tfpt_4_frontier`	**Frontier.** The honest status of η_B, m_p/m_e, Koide, dark matter and quantum gravity — the genuine handle, the precision, and what is *not* forced onto the ladder.
`tfpt_horizon_readouts` **(App. H)**	**Appendix H — horizon unit system (reframe, not new physics).** One seam constant c_3=$\frac{1}{8\pi}$ as the universal horizon thermal code: Hawking/de Sitter/Unruh, BH thermodynamics, Page, scrambling, Nariai bound, v_{GW}=c — standard physics in seam units, two compiler fingerprints (1920=$\lvert W(D_5)\rvert$, $\lvert\mu_4\rvert$=4).

The five formerly "research-level" problems are now *closed, reduced or typed* inside the documents where they belong — the group-level closure $(\text{Spin}(10) \times SU(4))/\Delta\mathbb{Z}_4$ in document 1 (closed), and the H2 splitting type, the Starobinsky scalaron mass, the θ_{12} texture and the quark c in document 2 (the first two closed; the quark *ratios* now closed combinatorially (v49/v71), only the *absolute* scale a U_{point} anchor; θ_{12} the seam-misalignment residual). Five companion files (plus this introduction) instead of a dozen, each result where it logically lives.

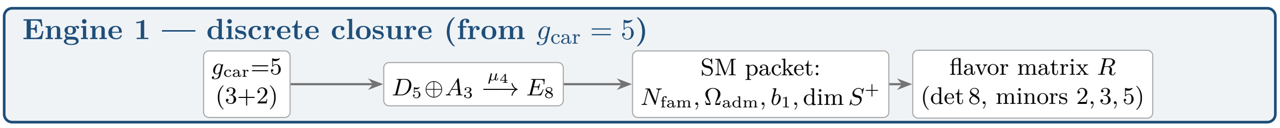
2 The architecture at a glance

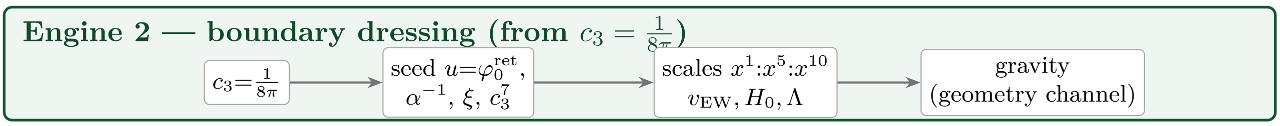
The diagram below shows the whole pipeline: *two* inputs on the left, the discrete compiler in the middle, the physics on the right. Everything in between is a consequence.



Key statement (the loop closes): previously D_5, A_3 and E_8 sat *as given* Lie structures in the background. Now they are *intermediate products* of the two inputs — *and the red back-channel is the new part*: the E_8 closure feeds back and *fixes the inputs.* $g_{\rm car} = 5$ is the unique rank-fill/Coxeter/Pascal solution ($2^{g-1} = \binom{g}{0}+\binom{g}{1}+\binom{g}{2}$), and the integer 8 in $c_3 = \frac{1}{8\pi}$ is rank $E_8 = h(D_5) = \det R$. **Inputs and output form an overdetermined fixed point** — the structure does not "prove itself"; it has fewer degrees of freedom than independent consistency conditions. The only thing *not* produced by the loop is the continuous π (Möbius/Gauss–Bonnet), the lone irreducible primitive. So it is not a one-way arrow chain but a closed, overdetermined self-consistency loop.

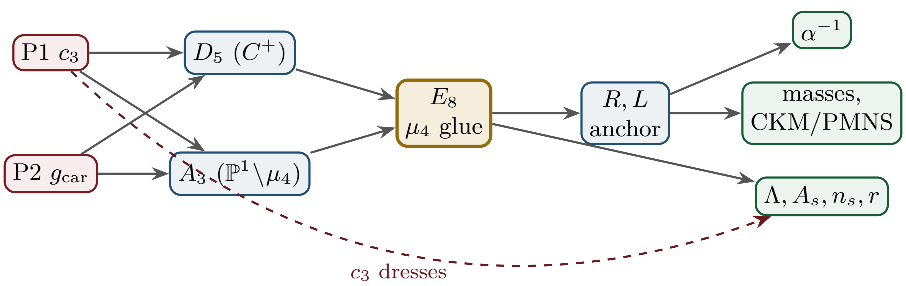
The master story is two engines. Read from the two axioms, the whole theory factorises into exactly *two* engines — a *discrete* closure and a *boundary* dressing. Gravity is not a third block; it is the geometry channel of Engine 2.





3 The dependency DAG and the proof ledger

The whole theory is a directed acyclic graph: two axioms at the source, the compiler in the middle, the observables as sinks. The DAG makes the attack surface explicit — each arrow is a named theorem or a declared input, and each node fails in exactly one way.



Node	Definition	Inputs	St.	Output	Failure mode
P1	RP boundary kernel, c_3=$1/(8\pi)$	— (axiom)	[A]	c_3	no reflection-positive seam
P2	5-slot carrier g_{car}=5	— (axiom)	[A]	carrier 3+2	wrong family/charge lattice
D_5	C^+ even-Hamming = half-spinor 16	P2	[I]	dim S^+=16, Y	group/matter mismatch
A_3	$\mathbb{P}^1 \setminus \mu_4$ four-puncture geom.	P1	[I]	N_{fam}=3, $\mathbb{Z}_6$	wrong multiplicity
E_8	μ_4 glue: disc=$\mathbb{Z}_4$, q-sum= 2	D_5, A_3	[I]	240, 248	not even-unimodular
R, L	residue + winding matrix, anchor $(1, 1, 2)$	A_3	[I]	det 8, minors $(2, 3, 5)$	wrong D_6 branch
α^{-1}	root of $F_{U(1)}(\alpha)$=0	c_3, M=41	[N]	137.0359992	no/second root (excl.)
masses	φ_0^{ret}-ladder $\lambda_Y^L \Lambda$	R, L, c_3	[I]/[P]	9 masses, mixings	hierarchy mismatch
θ_{12}	TBM + seam ε=c_3	R, L	[N]/[P]	0.307	seam-misalign. lemma fails
Λ, A_s	seam det. / R^2 scalaron	c_3	[P]	Λ, A_s, n_s, r	ambient RP fails

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

Reading the table top-to-bottom is reading the theory: two red axioms in, everything else a consequence with exactly one failure mode each.

The compact status formula (three honest layers)

$$\underbrace{\text{compiler}}_{\text{closed}} \;\Big|\; \underbrace{\text{admissible IR physics}}_{\text{conditionally closed (RP, gap)}} \;\Big|\; \underbrace{\text{strict physical TOE}}_{\text{open}}$$

TFPT 5.0 closes the discrete compiler, the algebraic Standard-Model readout, the EM fixed point, the admissible gapped IR sector and the $R + R^2$ spectral-action shadow. It does *not* yet certify a strict physical TOE. The remaining items have now collapsed sharply: the R-bridge amplitude normalisation U_{point} *reduces to* v_{geo} (Gate 1 closed, v75 — the same anchor as $1/G$), and the ambient metric measure is *holographically reduced* to a finite seam-*boundary* measure (Gate 2 tier B, v76) — which is in turn the rigorously-constructed $(E_8)_1$ lattice net (v77), so G_{metric} is imported into existing RCFT rigor rather than built anew. So the only genuine residuals are *one* dimensionful scale v_{geo} — the *dimensional-analysis floor* that no theory escapes (v78), shared by flavor and gravity — the boundary net identification, and the named QFT/cosmology transfer interfaces (F_{frontier}).

Two points that are *not* in the residual (closed, but easy to mis-list as open). *(i) Parameter-freeness is a theorem under the named hypotheses:* the boundary transport is gapped ($\Delta = 6 \log \frac{3}{2} > 0$), so by Perron–Frobenius its iteration has a *unique* attracting fixed point — the constants are *selected*, not tuned ([I]/[L] for the attractor, [`verification/v56_unique_attractor.py`]; the *physical identification* of the transport operator stays [P]), reinforcing the static bootstrap web ($g_{\text{car}} = 5$ forced three ways *given* the E_8 closure rank — a consistency loop, see red team Target B — [`verification/v6_bootstrap.py`], [`verification/v14_carrier_uniqueness.py`]). *(ii) Newton's constant is not an* independent *input:* fixing the physical Λ branch ($(8\pi)^2\delta_{\text{top}} = \frac{3}{4\pi^2}$) pins $\bar{M}_{\text{Pl}}$ *relative to* Λ, so G_N is read out *after* the branch choice + *one* dimensionful induced-gravity anchor ([I]/[P], [`verification/v60_lambda_metrology_branch.py`]), not predicted from pure numbers. What stays irreducible is then just π, the discrete E_8-closure fixed point (self-consistent, *not* a free axiom; the bootstrap of `tfpt_3`), and *one* dimensionful scale v_{geo} — and that single scale is shared by *both* the flavor sector ($U_{\text{point}} \to v_{\text{geo}}$, v75) and gravity ($1/G$, v68): the two [*A*] anchors are *one* (dimensional analysis; no metre from pure mathematics).

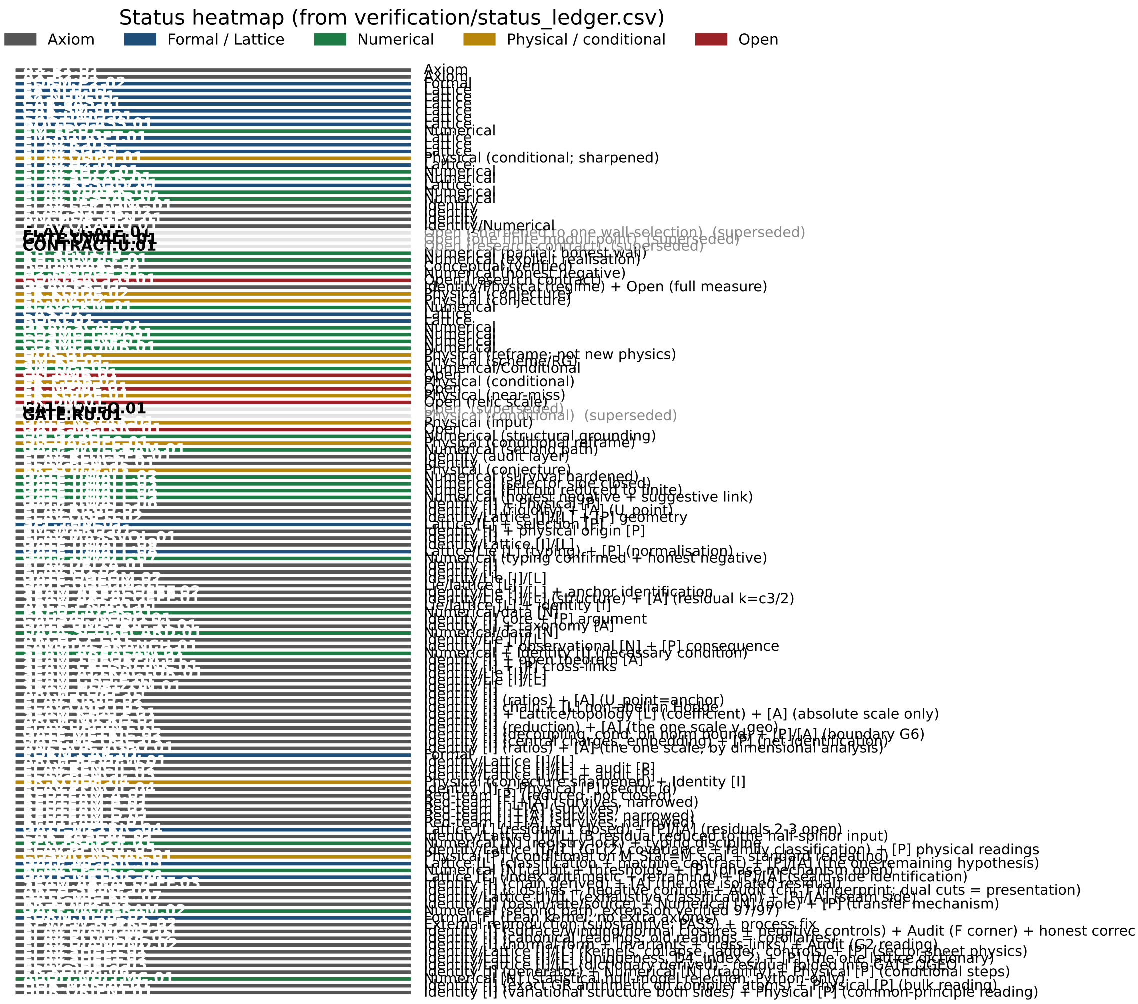


Figure 1: Status heatmap, generated directly from the single source of truth `verification/status_ledger.csv`. Grey = axiom, blue = formal/lattice, green = numerical, gold = physical/conditional, red = open. Each row carries a claim ID (`E8.GLU.01`, ...) that resolves to a ledger row with its dependencies and verification script. The ledger is *versioned*: each row has `active` and `canonical_status/supersedes` fields, so claims that were later superseded (e.g. the early "flavor gate open" rows, now *ratios closed* via `FLAV.RIGID`) are marked inactive and drawn dimmed — the canonical status is the only one read as current.

The hard reduction: everything open is two research gates plus typed interfaces

After the compiler closure, the entire residual is

$$\boxed{\text{Rest} = (U_{\text{wall}}) \ \oplus\ (G_{\text{metric}}) \ \oplus\ (F_{\text{frontier}})}$$

— one parabolic wall-selection, one quantum-gravity measure, and a set of deliberately typed QFT/cosmology interfaces. *Sharper aliases* (current state of the reductions; the historical gate names are kept for ledger continuity): $(U_{\text{wall}}) \to v_{\text{geo}}$ — the flavor ratios are *closed* (Plücker/Readout Rigidity, v49/v71/v75), what remains is *one dimensionful unit*, not a flavor gap; $(G_{\text{metric}}) \to G_{\text{net}}$ — reduced to a single boundary-net theorem, *"the seam–Calderón net is the index-4 μ_4 simple-current extension of the carrier net"* (holomorphy then follows, v89/`GATE.METRIC.06`); since the QBL chain (v108–v113) the *carrier choice rides on the same theorem*: the certified scalar kernel exists iff it pairs the sheets (v110), the certified channel counts the code identically (v112), pair transport is minimally complete (v111), and the one quasi-free kernel — rank $= c$: 5 carrier block, 8 seam hull — is the defining datum of the free net the gate already posits (v113), so closing G_{net} also closes the carrier selection [L]; $(F_{\text{frontier}}) \to F_{\text{transfer}}$ — named QFT/cosmology *transfer* interfaces (Koide source→pole, η_B Boltzmann, axion relic, m_p/m_e), never compiler powers. The status card:

Item	Status	Reading
parameter-freeness (self-consistency)	[I]/[L]	gapped transport ⇒ *unique* Perron–Frobenius attractor (v56); bootstrap web g_{car}=5 forced (v6/v14)
Newton constant G_N	[I]/[P]	not *independent*: branch pins $\bar{M}_{\text{Pl}}$ rel. to Λ (v60); readout after branch + one dim. anchor
ratio $c_u/c_d = \frac{55}{117}$	[I]	$g_{\text{car}}\lVert \text{Pl}_{\mathbf{1},a}(K)\rVert_1/(N_{\text{fam}}^2\Delta_Q)$; rigid on the derived stratum (v49/v71), no solve
(U) flavor gate	[A]$\to v_{\text{geo}}$	**Gate 1 closed**: $U_{\text{point}}\to v_{\text{geo}}$ (ratios + Grand Mass Volume, v75); same anchor as $1/G$
$R+R^2$ gravity	[I]/[P]	covariant $f(R)$ field equation closed in regime; heat-kernel grounded (G2)
physical QG sector	[F]/[P]	gap-decoupled admissible measure (Δ_{eff}=1.648>0)
ambient QG measure	[A](reduced)	**Gate 2 tier B**: holographically reduced to a finite seam-*boundary* measure (v76); IR tier closed (decoupling); strict-TOE cert. open (G_{metric})
Koide	[P]	$Q_{\text{src}} + \varphi_0/24$ as source→pole transfer conjecture
axion DM	[P]/[A]	$f_a = M_{\text{scal}}/128$ as a scenario
η_B	[P]	readout closed; leptogenesis Boltzmann solve missing
m_p/m_e	[A]	cross-sector QCD/EW ratio
$N_\star$	[P]	reheating input, $N_\star \simeq 57$ (continuous)
$a = (1, 1, 2)$	[L]/[P]	forced by $4 = 2 + 1 + 1$, conditional on the carrier
carrier selection g=5 (QBL)	[I]/[L] + [P]	theorem chain v108–v113: sheet-odd kernel, self-counting channel, transport degree 2, one quasi-free kernel (rank $= c$); residue = the G_{net} premise itself — gate closes ⇒ carrier [L]
E_8 Lean	[F]→[L]	hardening, not a blocker (script-certified)

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

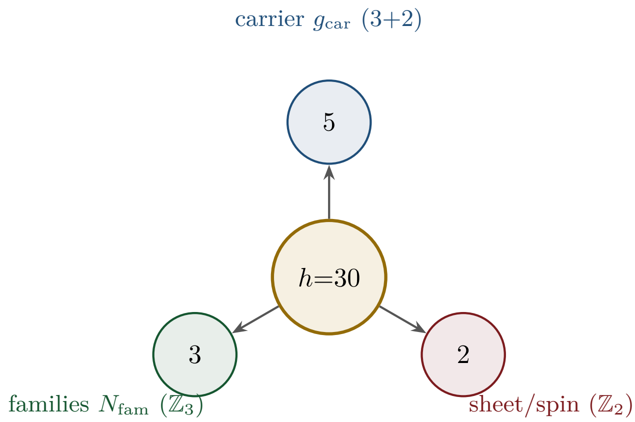
Only *two* research gates remain — (U_{wall}) and G_{metric}; the frontier items are typed interfaces, not structural holes. Of these, (U_{wall}) is the only genuine *physics* gate left: G_{metric} is now heat-kernel grounded (G2: the spectral action gives $R + R^2$ structurally) and *gap-decoupled* (G5: Δ_{eff}=1.648>0 isolates the closed admissible sector from the un-built ambient), so its residual — the ambient projective limit (G6) — is a *strict-TOE completion target*: its absence does not affect the bounded IR claim, but it does block certification as a strict physical TOE. Both are written up as numbered *research contracts* (lemma chains + closing theorem + machine-certifiability) in the companion `tfpt_research_contracts`, with priority (U_{wall}) first. [`verification/v36_spectral_action_g2.py`]

Reduction in one line

The number of *structural* assumptions is **two axioms**. Everything else ($N_{\text{fam}}, \Omega_{\text{adm}}, b_1, \alpha^{-1}$, the flavor matrix, the E_8 glue, the scale grammar) is a *consequence*. *And even those two tighten:* the bootstrap note shows $g_{\text{car}} = 5$ and the integer 8 in c_3 are *overdetermined E_8-closure fixed points* (rank-fill, Coxeter-match, integer-glue/norm all give 5; $8 = h(D_5) = \operatorname{rank} E_8 = \det R = \varphi(30)$), leaving *one* genuine continuous primitive: π from the Möbius boundary. So the *discrete* core has no freely tunable fundamental numbers — the bootstrap is *not creation from nothing* (two inputs remain), but an overdetermination of those inputs. (This concerns the discrete core only; dimensionful masses, the QG measure and frontier items remain typed-open.)

4 The compiler: why $30 = 2 \cdot 3 \cdot 5$

The Coxeter number of E_8 is $h = 30 = 2 \cdot 3 \cdot 5$. Exactly these three prime factors are the three discrete atoms of the theory — that is the point of the compiler:

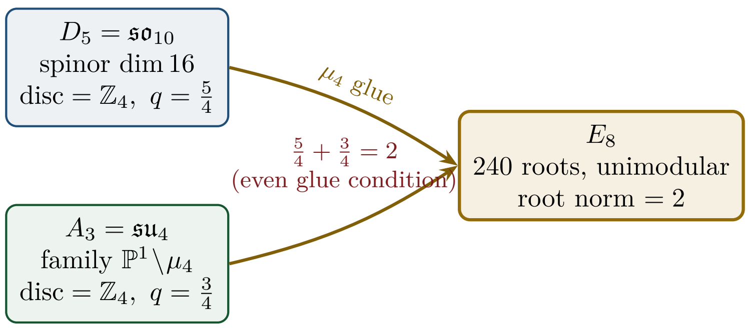


From this the compiler reads, among others:

- $\dim S^+ = 1 + 10 + 5 = 16$ (one generation) as the even exterior algebra of the 5-slot carrier — the same Pascal row $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$ appears in family, action ladder and the E_8 root count. [I]
- $|R(E_8)| = 16 \cdot 5 \cdot 3 = 240$ and $\dim E_8 = 240 + (5 + 3) = 248$. [I]
- the $\mathbb{Z}_{30}$ Coxeter element $\leftrightarrow$ the cyclotomic polynomial Φ_{30}; the corner rotation $z \mapsto iz$ on $\mathbb{P}^1 \backslash \mu_4$ is the A_3 Coxeter element. [I]

5 The μ_4 glue: how E_8 is really built

The decisive, now *proved* step: D_5 and A_3 have the *same* discriminant group $\mathbb{Z}_4$, and their discriminant-form norms are exactly two TFPT constants which add up to the E_8 root norm.



Glue theorem (proved)

$\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4$, glue index $|\mu_4| = 4$, and $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2 =$ the E_8 root norm. Hence $E_8 = (D_5 \oplus A_3) + \mu_4$-glue is a *closed lattice-theoretic* construction — not a blind "positing of 248". [L]

6 How derivation now works: the derivation map

What can one concretely *compute* with $\{c_3, g_{\rm car}\}$? The following map summarises what is now a consequence rather than an assumption.

Sector	How it now arises	Status
gauge group / families	carrier 3+2, $\dim S^+=16$, $N_{\text{fam}}=3$, $\Omega_{\text{adm}}=48$	[I]
hypercharge	roots of the charge polynomial $bsY^2+(s-b)Y-1$	[I]
$U(1)$ index b_1	$b_1=\dfrac{g_{\text{car}}\,2^{g_{\text{car}}-2}+1}{10}=\dfrac{41}{10}$	[I]
fine structure α^{-1}	unique root of $F_{U(1)}(\alpha)=0$	[I]/[N]
flavor matrix	transport theorem (H1 proved, H2/H3 force the matrix)	[I]/[P]
electroweak scale	$v_{\text{EW}}\sim e^{-\alpha^{-1}/5}$ (carrier divides α^{-1} by 5)	[I]
cosmological constant	$\Lambda\sim e^{-2\alpha^{-1}}$ (density quotient; see typing note)	[I]/[P]
Hubble scale	$H_0\sim\sqrt{\Lambda}$	[P]
inflation amplitude	$A_s=\dfrac{N_\star^2}{24\pi^2}c_3^7$ in the R^2 branch	[P]
E_8	$D_5\oplus A_3+\mu_4$-glue	[L]

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

Notable: the scale grammar is *one* exponential engine. The same number $\alpha^{-1}\approx 137$ generates the electroweak scale (divided by 5 = carrier), the cosmological constant (times 2) and — via the square root — the Hubble scale. One constant becomes many orders of magnitude.

Typing note on Λ (a deliberate danger zone). "Λ" denotes *three* distinct objects that must not be conflated: (i) the seam-determinant exponent $e^{-2\alpha^{-1}}$ (an exact [I] readout), (ii) the dimensionless density quotient $\rho_\Lambda/\bar{M}_{\text{Pl}}^4$ (which carries a prefactor *branch*, hence [P]), and (iii) the observed curvature, to which (ii) is *anchored*, not predicted. The exponential is exact; the absolute dark-energy density remains a typed cosmology interface (full disambiguation in `tfpt_2_standard_model`).

Cosmology is the Pascal action grammar of the carrier — not a separate module

With $x=v_{\text{geo}}/(5\beta_{\text{rad}}^2\bar{M}_{\text{Pl}})$ the cosmological readouts are a *power tower* on the five-slot carrier graph K_5:

$$\boxed{\frac{H_0}{\bar{M}_{\text{Pl}}}=x^5R_H,\qquad \frac{\rho_\Lambda}{\bar{M}_{\text{Pl}}^4}=x^{10}R_\Lambda},\qquad (1,5,10)=\binom{5}{0},\binom{5}{1},\binom{5}{2}.$$

EW is the unit x^1, Hubble is the product over the $\binom{5}{1}=5$ vertices, Λ is the product over the $\binom{5}{2}=10$ edges — the *same* Pascal triple that reads one generation, the action ladder and the E_8 root count $16\cdot 5\cdot 3=240$. "Hubble = slot product, Λ = pair product." [I]

The entire fermion spectrum in one formula

Masses, Yukawa, CKM, PMNS and neutrinos all follow from *one* master formula with *one* seed:

$$\hat{m}_{f,j}=\frac{v_{\text{geo}}}{\sqrt{2}}\lambda_Y^{L_{f,j}}\Lambda_{f,j},\qquad \lambda_Y=\sqrt{\varphi_0^{\text{ret}}(1-\varphi_0^{\text{ret}})},\qquad \varphi_0^{\text{ret}}=\tfrac{1}{6\pi}+\tfrac{3}{256\pi^4}.$$

The word-lengths $L_{f,j}$ are the *fixed residue matrix of the compiler*, whose characteristic polynomial $t^3-9t^2+10t-8$ carries only compiler numbers ($\text{tr}=9=N_{\text{fam}}^2$, $\det=8=h(D_5)$, principal 2-minors 2,3,5 with product $30=h(E_8)$). Through the φ_0^{ret}-ladder $\hat{m}=\frac{v_{\text{geo}}\pi}{\sqrt{2}}c\,(\varphi_0^{\text{ret}})^k$ all nine word-lengths are fixed and the *charged-lepton* amplitudes are closed; the *quark* mass *ratios* are closed too (integer Plücker readouts on the derived selector stratum, v49/v71), with only the *absolute* amplitude scale reducing to the (U_{wall}) anchor. **SM status:** the discrete packet, the dimensionless flavor skeleton, the charged-lepton source masses and the quark *ratios* are closed; the *absolute* quark scale and the full PMNS completion are reduced/conditional; $m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s$ are RG scheme-layer (m_H rests on the closed UV quartic $\frac{1}{16\pi^2}$); and the solar angle is *realized by a TFPT texture* and reduced to the seam-misalignment lemma ($\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$; see below). *(Full derivation: document 2,* `tfpt_2_standard_model`*.)*

7 Physical, geometric, topological: why the architecture is plausible

The closure works on three levels at once: *physically*, α^{-1} is the fixed point of *one* equation and one generator drives many scales; *geometrically*, the C^+ spinor and the puncture geometry $\mathbb{P}^1\backslash\mu_4$ produce D_5 and A_3, glued to E_8; *topologically*, the common discriminant $\mathbb{Z}_4$ (the μ_4 winding) forces the glue and Φ_{30}/Coxeter drives the sectors — with exactly 2 axioms and the rest a consequence.

Why the compiled architecture is plausible

It is (i) *parsimonious* (few independent assumptions $\Rightarrow$ low overfitting risk), (ii) *rigid* (the glue condition $5/4 + 3/4 = 2$ and the discriminant $\mathbb{Z}_4$ leave little freedom), and (iii) *checkable* (machine-checkable E_8 appendix, explicit fixed-point equation for α^{-1}). More explained from less — exactly the criterion that makes a theory more probable.

8 What role E_8 now plays exactly

E_8 is no longer the starting point but the *output format*:

- **Bookkeeping:** E_8 is the unimodular container in which carrier (D_5) and family (A_3) fit together consistently. Its numbers $240, 248$ are carrier traces, not inputs.
- **Consistency test:** unimodularity ($\det 1$) is a hard condition; it closes only because the discriminant-form norms $5/4$ and $3/4$ (both already present in TFPT) add up to 2. E_8 thus tests the internal coherence of the two atoms.
- **Not an end in itself:** E_8 does not explain "everything"; it is the smallest even unimodular hull of the datum $D_5 \oplus A_3$. The physics sits in the *two inputs*, not in E_8.

New — the secondary atlas. Beyond that, E_8 is an *audit container*: each maximal subalgebra projects a TFPT module. Document 3 (`tfpt_3_e8_audit_bootstrap`, Part I) shows the seven slices of 248 explicitly and labelled. The strongest new hit:

$E_6 \times A_2$ reads the flavor matrix

$$248 = \|R\|_F^2 + \det R + 2\,(\mathbf{1}^\top R\,a)\,N_{\text{fam}} = \underbrace{78}_{\dim E_6} + \underbrace{8}_{\dim A_2} + 2 \cdot 27 \cdot 3,$$

with $\|R\|_F^2 = 78 = \dim E_6$, $\det R = 8 = \dim A_2 = h(D_5)$. The discipline rule is: *every load-bearing number must appear in at least one E_8 projection* — this makes the structure falsifiable rather than numerological. (Audit level; all identities machine-verified.)

9 What this means for the theory-of-everything question

The state of the five questions — closed, reduced or typed

1. **Flavor residual — ratios closed; only the absolute scale is the wall-selection.** Mehta–Seshadri supplies the equivalence-class framework (stable parabolic $\leftrightarrow$ unitary representation); the algebraic selector R, the Q spectrum and the splitting type are fixed (now *derived*: $\det R = 8$ lattice, $\mathrm{Spec}(Q_+)$ from D_4, v69/v70), and the charged-lepton amplitudes are closed. The quark mass *ratios* ($c_u/c_d = \frac{55}{117}$ etc.) are then *constant* on this discrete selector stratum (Readout Rigidity, v49/v71) — integer Plücker readouts, no transcendental solve. Only the *absolute* amplitude scale reduces to the selected *polystable* point on the parabolic stability wall (U_{point}, an anchor; Research Contract 1). [I]/[A]

2. **Gravity — exact in the R^2 regime.** Spectral action $\to R{+}R^2$, BRST/Hodge kills the gauge directions, FRW reduction $\to$ Starobinsky; the only remainder is the *controlled* low-curvature truncation ($R^{n\geq 3} \to R^2$, valid for curvature $\ll M^2$). [I] in the regime

3. **Strong CP — decoupled.** $\theta_{\text{eff}}{=}0$ follows from three structural facts (γ_5-Hermiticity, polar structure, sheet involution) + reflection positivity — *without* a mass gap. [I] (given RP)

4. **Full dynamics — the admissible transfer model is closed.** The mass gap is the *explicit* discrete gap $6 \log \frac{3}{2}$ of the C_6 transfer operator (eigenvalues $(k/3)^6$), which supports the OS reconstruction path on the admissible sector; the continuum and metric-coupled theory still require the stated RP, cluster and limit assumptions. [I] (transfer model) / [P] (continuum)

5. **Axioms P1/P2 — P2 machine-verified.** The Lean proof builds cleanly (`1886 jobs`, 0 `sorry`, only kernel axioms); the P2 algebra is a theorem, P1 a typed interface. They remain declared inputs. [I]/[A]

In other words: the theory has shrunk from "many open audit points" to "*standard steps* (cluster expansion, Fredholm, truncation bound) + two declared axioms". The five questions are now *closed, reduced or typed* rather than open-ended — a qualitative jump in discipline, not a claim of completion. *(Details and proof chains: document 2, `tfpt_2_standard_model`, Part III.)*

The TOE-completion ledger — what is done, and what genuinely remains

A theory-of-everything must close the discrete structure, the dimensionless constants, the mass spectrum, gravity, and the quantum measure. The current state, graded honestly:

TOE piece	**Status**	**If not complete: what closes it**
gauge group, N_{fam}, hypercharge, b_1	[I]	— complete
$E_8 = (D_5 \oplus A_3) + \mu_4$, group closure	[L]	— complete
bootstrap $(g,\mu)=(5,4)$, "8" in c_3	[L]	— complete (reverse glue $\mu^2-5\mu+4=0$)
fine structure α^{-1}	[N]	— **existence+uniqueness proved**; value is a numerical fixed point (ledger `EM.FP.01`), Ward origin [P]
fermion masses (φ_0^{ret}-ladder)	[I]/[A]	leptons exact; quark *ratios* closed (Plücker, v49/v71); absolute scale = anchor
mixings $\theta_{13}, \theta_{23}, \theta_{12}$	[N]/[P]	θ_{12} cond. derived; θ_{23} maximal
flavour matrix / H2	[I]	up to Mehta–Seshadri unitarity (U), a Paper-3 input
inflation A_s, n_s, r, scalaron M	[N]	parameter-free given $M_{\mathrm{Star}}=M_{\mathrm{scal}}$
genuinely open — the actual TOE remainder		
dim. EW/QCD masses	[P]	standard RG threshold matching (no new physics)
QFT measure (admissible sector)	[I]	**closed: RP from cusp spectrum, OS reconstruction**
ambient QFT + full gravity (metric sector)	[A]	*not constructed here* (frontier doc); conditionally reduces to one named hypothesis (ambient RP) on the relative spectral-action measure
analytic boundary origin of π ($c_3=\frac{1}{8\pi}$)	[P]	**hardened:** $c_3=1/(\lvert\mathbb{Z}_2\rvert \oint_{S^2} K) = 1/(8\pi)$ **(Gauss–Bonnet)**; "8"= rank E_8; π stays the one primitive
η_B, m_p/m_e, Koide, dark matter	[A]	frontier handles only (see document 4)

Marker key: [I] exact identity · [L] Lie/lattice theorem · [F] formalised · [N] numerical fixed point · [P] physical/conditional · [A] axiom / open.

The honest bottom line: the discrete core, the dimensionless constants, the *mass-hierarchy exponents* and the mixings are *closed*; the nine masses follow from one φ_0^{ret}-ladder (lepton rationals exact, quark $O(1)$ digits still a conditional finite evaluation). The *admissible-sector* QFT measure is closed (a finite gapped transfer system, RP from the cusp spectrum, OS-reconstructed); the *ambient/metric-sector* measure — i.e. *full quantum gravity* — is *not* constructed here and stays [A] (frontier doc), reducing conditionally to a single named hypothesis (ambient reflection positivity). The seam seed is *hardened*: $c_3 = 1/(\lvert\mathbb{Z}_2\rvert \oint_{S^2} K\,dA) = 1/(8\pi)$ is the one-sided Gauss–Bonnet normalisation of the seam sphere, with the discrete "8" equal to rank E_8 and only the Gauss–Bonnet π left as the single irreducible primitive. What remains for a full TOE is then the ambient-RP hypothesis (full QG), plus the standard-RG dimensionful masses and the honest frontier items. No load-bearing discrete parameter remains free, and π is the lone continuous primitive.

10 Does this make the theory more probable?

Plausibility balance

Pro: (a) two inputs instead of many; (b) α^{-1} to 2.9×10^{-10} (CODATA-2022, 1.9σ) as a *fixed point*, not a fit; (c) E_8 constructed, not postulated; (d) one compiler generates several sectors ⇒ fewer degrees of freedom, more predictive power.

Contra/open: what remains are only *standard steps* (cluster expansion in the thermodynamic limit, the Fredholm property, the truncation bound) and the two declared axioms P1/P2; on the SM side the dimensionful EW/QCD masses ($m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s$) sit on the RG scheme layer. The solar angle, previously the one open SM number, is now *conditionally derived* ($\frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2}$, modulo the seam-misalignment lemma). A final verdict "it maps reality" requires this short list plus the near-term experimental tests (below).

Net: the parsimony (Kolmogorov-like compression: many numbers from little) and the high precision of individual closures raise the a-posteriori plausibility substantially — without hiding the open points.

11 Predictions for running and upcoming experiments

Because *one* generator now sits behind the numbers, the theory makes a set of concrete, falsifiable statements that running or near-term experiments are actively testing. Each row gives the TFPT value, the *new* derivation behind it, the experiment, and an honest status.

Observable	TFPT value & derivation	Experiment (running / upcoming)	St.
solar angle $\sin^2\theta_{12}$	$\frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2} = 0.3067$; TBM + seam misalignment $\varepsilon = \frac{3}{4}\varphi_0^{\rm ret} \approx c_3$ (cond.)	**JUNO** (taking data since Aug 2025; sub-percent θ_{12}) — the sharpest live test	[N]/[P]
reactor angle $\sin^2\theta_{13}$	$\varphi_0^{\rm ret} e^{-5/6} = 0.0231$; seed × carrier-trace $e^{-\gamma}$, $\gamma = \frac{5}{6}$	Daya Bay / RENO / JUNO (PDG 0.0220)	[I]
atmospheric $\sin^2\theta_{23}$	$\approx \frac{1}{2}$ ($\mu\tau$-symmetric limit; octant not selected)	NOvA / T2K / **DUNE** (octant ambiguity, NuFIT 6.0)	[P]
fine structure α^{-1}	137.0359992 as the unique root of $F_{U(1)}(\alpha)=0$ (no fit, no second branch)	CODATA-2022 137.035999177(21): dev. 2.9×10^{-10} from the recommended adjustment (1.9σ of its uncertainty)	[I]/[N]
tensor ratio r	$r = \frac{12}{N_\star^2} = 0.0040$ ($N_\star$=55); R^2 attractor, $M = c_3^{7/2} \bar{M}_{\rm Pl}$	now $r<0.036$ (BICEP/Keck BK18); **CMB-S4** ($\sigma_r \le 5\times10^{-4}$, obs. ∼2033)	[P]
tilt n_s	$n_s = 1 - \frac{2}{N_\star} = 0.964$ (same attractor)	Planck (0.965); CMB-S4 sharpens	[P]
amplitude A_s	$\frac{N_\star^2}{24\pi^2} c_3^7 = 2.0\times10^{-9}$, parameter-free	Planck (2.1×10^{-9})	[P]
neutron EDM	$\theta_{\rm eff} = 0 \Rightarrow d_n$ essentially null (structure + RP)	**PSI nEDM** / SNS (running)	[I]
$0\nu\beta\beta$ / ordering	Majorana branch; normal-ordering preference	**LEGEND / nEXO**, DUNE, KATRIN	[P]
flavor invariants	$\det R = h(D_5) = 8$, principal minors $(2, 3, 5)$, $\chi_R = t^3 - 9t^2 + 10t - 8$ (exact)	any future CKM/PMNS global fit must satisfy these	[I]
H_0 vs. Λ	$v_{\rm EW} \sim e^{-\alpha^{-1}/5}$, $\Lambda \sim e^{-2\alpha^{-1}}$, $H_0 \sim \sqrt{\Lambda}$: one engine	SH0ES / **DESI** / Planck (Hubble tension)	[P]

*Marker key: **[I]** exact identity · **[L]** Lie/lattice theorem · **[F]** formalised · **[N]** numerical fixed point · **[P]** physical/conditional · **[A]** axiom / open.*

The two decisive near-term tests

(1) JUNO and θ_{12}. JUNO is *taking data since August 2025* and targets sub-percent $\sin^2\theta_{12}$. TFPT commits to the conditional $\frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2} = 0.3067$ (seam $\varepsilon = c_3$). A measured central value away from ~ 0.307 at high significance falsifies the seam mechanism — a *live* test.

(2) r. The R^2 scalaron with M fixed by c_3^7 predicts $r = 0.0040$ ($N_\star$=55), already below the current bound $r < 0.036$ (BICEP/Keck BK18) and within reach of CMB-S4 ($\sigma_r \le 5 \times 10^{-4}$, first observations ∼2033–34). A detection of $r \gtrsim 0.01$ would be incompatible with the R^2 branch on which $M_{\rm Pl}$ and A_s rest.

Structural (non-experimental) checks. Independent of any apparatus, the machine-checkable E_8 appendix (all 240 roots, Gram matrices, discriminant groups) lets anyone verify that the two

TFPT atoms really generate E_8; and the discipline rule "*every load-bearing number appears in at least one E_8 projection*" (atlas note) turns the number stock into a falsifiable raster rather than numerology. The numerology charge itself is additionally *quantified* by an explicit null model: exhaustively enumerating a declared formula grammar (provably containing every scored TFPT formula) against conservative data windows gives a joint look-elsewhere-corrected null probability $\leq 10^{-30.7}$ that a random theory of equal formula complexity reproduces the scorecard — a statistical statement conditional on the declared grammar, not a claim of certainty **[N]** [`verification/v100_numerology_null_mc.py`].

Freeze file: committed kill criteria

To prevent post-hoc rescue, the predictions above are frozen with explicit kill criteria in the machine-readable registry `verification/freeze_file.csv`. The decisive entries:

- **solar** $\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} \approx 0.3067$ — a JUNO central value clearly away from 0.307 kills the seam-misalignment mechanism;
- **tensor** $r \approx 0.0040$ — any robust $r \gtrsim 0.01$ kills the R^2 branch carrying M_{Pl}, A_s;
- **ordering** normal, small $m_{\beta\beta}$ — inverted ordering or large $m_{\beta\beta}$ kills the Majorana branch;
- **strong CP** $\theta_{\text{eff}} = 0$ — a solid nEDM signal above SM background falsifies the structural cancellation;
- $w = -1$ — a robust $w \neq -1$ kills the single-engine dark-energy readout.

The proton/electron ratio is explicitly *not* claimed as a compiler power (only fails if mis-asserted as one).

Blind-prediction registry (frozen 2026-06-09, machine-enforced). Beyond the kill criteria, the *values* themselves are now frozen: `verification/predictions_frozen.json` registers every dimensionless prediction of record at 25 significant digits *before* JUNO / CMB-S4 / LiteBIRD / DESI decide, and [`verification/v84_frozen_registry.py`] re-derives each decimal from the two axioms on every suite run (formula↔value lock; covered by `manifest.sha256`; ledger `REG.FREEZE.01`). In particular the registry admits exactly *one* θ_{12} prediction of record (the seed $\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2} = 0.306747$) — the seam and non-linear variants are typed as derived variants of the same texture and can never be silently promoted afterwards. r and n_s enter only as *bands* over the declared external $N_\star \in [50, 60]$, never as point predictions.

What likely follows next

- the quark mass *ratios* are now closed combinatorially (integer Plücker readouts on the derived selector stratum, v49/v69/v71); the only residual is the *absolute* amplitude scale, an anchor (U_{point}) of the same nature as the one dimensionful scale — no new physics;
- further constants as carrier traces (analogous to $b_1 = 41/10$), since the compiler systematically produces traces over S^+;
- a sharper geometry↔gravity bridge once the Hodge/R^2 branch is lifted from **[P]** to **[I]**.

Conclusion

TFPT's sectors form an impressive collection of closures sharing common numbers. The compiler closure explains *why* they are the same numbers: a small discrete generator $\mathbb{Z}_{30} = 2 \cdot 3 \cdot 5$ builds, from two inputs $\{c_3, g_{\text{car}}\}$ via $D_5 \oplus A_3 + \mu_4$, the E_8 container and outputs the Standard Model, the constants and the scale grammar from there. The theory thereby becomes *more parsimonious, more rigid and more checkable.* Precisely nameable gaps remain — one geometric flavor realisation, the conditional gravity/QFT closures, and two declared axioms — but the path to closure is now short and concrete.

Appendix: computational verification (reproducibility)

Every claim in this document marked [I] (exact identity), [L] (Lie/lattice theorem) or [N] (numerical fixed point) is re-derived from the two axioms $\{c_3, g_{\rm car}\}$ alone by a small, self-contained Python suite in the repository folder `verification/`. The inline references `[verification/...]` throughout the documents point to the exact script; the table below is the master map.

Script	**What it machine-checks**
`v1_e8_glue.py`	glue theorem $E_8{=}(D_5{\oplus}A_3){+}\mu_4$: disc $=\mathbb{Z}_4$, $q(D_5){+}q(A_3){=}\frac{5}{4}{+}\frac{3}{4}{=}2$, $240{=}16{\cdot}5{\cdot}3$, $248{=}240{+}8$
`v2_carrier_pascal.py`	$g_{\rm car}{=}5$ unique Pascal solution; $16{=}1{+}5{+}10$, $N_{\rm fam}{=}3$, rank $E_8{=}8$, $\Omega_{\rm adm}{=}48$, $b_1{=}\frac{41}{10}$
`v3_em_alpha.py`	EM closure: unique root of $F_{U(1)}(\alpha){=}0$, $\alpha^{-1}{=}137.0359992168$, existence/uniqueness, inverse test, ablation
`v4_flavor_matrix.py`	residue matrix R: det $=8$, minors $\{2,3,5\}$, $\chi_R{=}t^3{-}9t^2{+}10t{-}8$, SNF diag$(1,1,8)$, $\lVert R\rVert_F^2{=}78$, $\sum L{=}40$
`v5_e8_cascade.py`	cascade $D_n{=}60{-}2n$: endpoints, exponent rungs $\to$ 240, IR tail 46080, variance 6552
`v6_bootstrap.py`	reverse glue $\mu^2{-}5\mu{+}4{=}0$; $g_{\rm car}{=}5$ three ways; $8{=}$ rank $E_8{=}h(D_5){=}\varphi(30)$
`v7_gravity_cosmo.py`	scalaron exponent 7, $M^2_{\rm scal}/\bar{M}^2_{\rm Pl}{=}c_3^7$, A_s, n_s, r, Ω_b, $\sin^2\theta_{12}$
`v8_horizon.py`	horizon code: $\frac{1}{2\pi}{=}4c_3$, $1920{=}\lvert W(D_5)\rvert$, S_{dS}, Page, $\beta_{\rm rad}{=}0.2424°$
`v9_neutrino_texture.py`	solar angle: explicit $\mu\tau$ Majorana texture $\to \sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\rm ret}}{2}$, $\theta_{23}{=}45°$, $\theta_{13}{=}0$
`v10_projection_involution.py`	Q, Σ algebra: $K{=}R{+}Q\Sigma$, $L{=}R{+}Q(I{+}\Sigma)$; χ_Q, χ_K; det ladder $3{\cdot}4{\cdot}8{\cdot}20{=}1920{=}\lvert W(D_5)\rvert$; $a^\top Ka{=}41$
`v11_unique_KQ.py`	K, Q are the *unique* nonneg-int matrices with their compiler budgets, spectrum and monotone rows (enumeration)
`v12_mass_generation_polynomials.py`	sector/generation polynomials of K and the anchor-block det ladder $(9,10,16,40)$
`v13_open_gates.py`	gate closures: $M{=}41{=}10b_1{=}\sum L{+}N_\Phi$; $Q_+ = A_3$ exponents, $Q_-^2{=}N_{\rm fam}$
`v14_carrier_uniqueness.py`	$g_{\rm car}{=}5$ unique ($N_{\rm fam}\in\mathbb{Z}^+$, rank $E_8{=}8$); split $(3,2)$ from $b{+}s{=}5, b^2{+}s^2{=}13$; $\mathrm{Tr}\,Y{=}\mathrm{Tr}\,Y^3{=}0$
`v15_bootstrap_classification.py`	$D_5{\oplus}A_3$ unique familyful cyclic-glue of E_8 (16-spinor + 3 families), glue $\mathbb{Z}_4$
`v16_solar_dual_anchor.py`	$a^\top R^{-1}{=}a^\top L^{-1}{=}(-\frac{1}{2},-\frac{1}{2},1)$; $\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\rm ret}}{2}$; full PMNS
`v17_hexagonal_resolvent.py`	finite resolvent $D_y^{-1}{=}(\sum y^{5-m}\delta^m U_6^m)/(y^6{-}\delta^6)$ (quark c backbone)
`v18_quark_yukawa.py`	quark source ratios $\frac{m_u}{m_d}{=}0.470085$, $\frac{m_c}{m_s}{=}13.61$, $\frac{m_t}{m_b}{=}40.80$ exact; lepton $c{=}(\frac{16}{7},\frac{4}{3},\frac{7}{6})$; full hierarchy
`v19_monodromy_moduli.py`	exact pole $z_\star{=}\frac{794-7\sqrt{9961}}{2187}$ (3^7); D_4 $SU(3)$ monodromy constructible but D_4-fixed locus positive-dim $\Rightarrow$ exact quark c reduce to (U)
`v20_lepton_c_derivation.py`	lepton c's derived: $\delta{=}\frac{1}{2}$ resolvent $c{=}\lvert\mu_4\rvert^w/(\frac{5}{4}{-}\cos\frac{r\pi}{3})\Rightarrow\frac{16}{7},\frac{4}{3}$; product $\frac{2^{g_{\rm car}}}{N^2_{\rm fam}}{=}\frac{32}{9}\Rightarrow\frac{7}{6}$
`v21_solar_product_quark.py`	solar coeff $= q(A_3){=}\frac{3}{4}$ (glue-norm, $q(D_5){+}q(A_3){=}2$), $\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\rm ret}}{2}$; product $\frac{2^{g_{\rm car}}}{N^2_{\rm fam}}$; quark law fails ($\frac{1}{7}$ vs 0.724)

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Script	What it machine-checks *(continued)*
`v22_open_gates_audit.py`	residual-gates audit contract: exact reduction of each open gate (A2,B3,B4,B5,B6,C7) machine-pinned (forced part vs. named residual)
`v23_anchor_generator.py`	anchor $a{=}(1,1,2)$: $e_k{=}(4,5,2)$, $c_3{=}\frac{1}{2e_1\pi}$; $p_n{=}2{+}2^n \Rightarrow 240, 248$, binary ladder; inputs $\to \{a,\pi\}$
`v24_quark_ratio_closure.py`	quark ratios $\frac{55}{117}, \frac{34}{47}, \frac{3}{26}$ from TFPT blocks; mass ratios match $< 0.03\%$; rational c gauge, Λ_q in $O(1)$
`v25_frontier_conjectures.py`	conjectures: Koide $Q_{\rm src}{+}\frac{\varphi_0^{\rm ret}}{24}{\approx}\frac{2}{3}$; axion $f_a{=}M_{\rm scal}/128{\to}m_a{\approx}23.8\,\mu$eV
`v26_flavor_frontier_unification.py`	the "11" is not uniquely forced ($\Rightarrow$ [P]); flavor frontier unifies to one (U) gate; cusp config on the parabolic stability wall
`v27_wall_representative.py`	explicit balanced wall rep $W_{\rm wall}$ (cols = perm$(0,1,2)$, rows $w{=}(2,1,1)$, pardeg 0); selector $\det R{=}8$, Spec$(Q_+){=}\{1,2,3\}$
`v28_gravity_fR.py`	$R{+}R^2$ closed: $f(R)$, $M{=}c_3^{7/2}\bar{M}_{\rm Pl}$ scalaron, $N_\star{=}57 \to n_s, r, A_s$; full metric measure open; $248c_3^2{<}6\log\frac{3}{2}$
`v29_research_contract_certs.py`	research-contract certificates: $C_U^{(1)}$ wall enumeration (1296→144→5 orbits; symmetry alone does not pick $W_{\rm wall}$) and $G5$ gap-dominance $2\\|V\\|{<}\Delta$
`v30_d4_character_variety.py`	$C_U^{(2)}$: full D_4-fixed $SU(3)$ character variety is positive-dimensional $\Rightarrow$ selector needs the $R(\rho)$ dictionary (H2; exact since v118–v122, residue = the R4′ selector establishment)
`v31_R_dictionary.py`	$R(\rho)$ characterized: case A alive (irreducible $\Rightarrow$ mixing); R integer but ρ-invariants continuous $\Rightarrow$ $R(\rho)$ is the transcendental non-abelian-Hodge map (residual = Hitchin solve)
`v32_rh_splitting.py`	RH route: $\sum_k U^k A_0 U^{-k}{=}4\,{\rm diag}(A_0)$ exact $\Rightarrow$ splitting $\mathcal{O}(-2){\oplus}\mathcal{O}(-1)^2 {\Leftrightarrow} {\rm diag}(A_0){=}(\frac{1}{2},\frac{1}{4},\frac{1}{4})$ (algebraic); $\prod M_k{=}I$ + R-bridge stay open (c_u/c_d not forced)
`v33_explicit_flat_bundle.py`	explicit valid ($U_{\rm wall}$) flat bundle (RH solve): cusp + $\mathcal{O}(-2){\oplus}\mathcal{O}(-1)^2$ + $\\|M_\infty{-}I\\|{\sim}10^{-9}$ ($\prod M_k{=}I$) + irreducible (case A)
`v34_h2_bridge_attempt.py`	H2 bridge: explicit per-puncture M_k (cusp, $\prod M_k{=}I$); honest negative — natural extraction $\neq$ lepton amplitudes, the $\Gamma^{\rm min}$ dictionary is the remaining input (c_u/c_d not obtained)
`v35_quark_qcd_boundary.py`	the open gates are the discrete↔continuous boundary: quark cross-ratio cleanness tracks generation/QCD sensitivity (compiler hands off to continuous physics)
`v36_spectral_action_g2.py`	G2: spectral action $\to R{+}R^2$ ($a_2{\sim}{-}R/3$, $a_4\|_{R^2}{\sim}R^2/72$); $M^2/\bar{M}_{\rm Pl}^2{=}6(4\pi)^2/f_0$, c_3^7 fixes f_0; gap-decoupling $\Delta_{\rm eff}{=}1.648{>}0 \Rightarrow$ admissible IR closed under RP, ambient (G6) = strict-TOE completion target
`v37_plucker_anchor.py`	anchor-plane Plücker apparatus: $\\|{\rm Pl}(K)\\|_1{=}11$, $\\|{\rm Pl}_R(K)\\|_1{=}26$, $\sum {\rm Pl}_R(L){=}60$; pencil $\det(K{+}xQ){=}3x^3{+}7x^2{+}6x{+}4$; $K_e{-}K_u{=}a$, $L_e{-}L_u{=}{\rm Exp}(A_3)$; lepton ring $c_e c_\tau{=}\|\mathbb{Z}_2\|c_\mu$; Koide $u{\to}\frac{53}{54}u$ ([P])
`v38_uwall_killswitch.py`	($U_{\rm wall}$) U2 kill-switch hardened: D_4-cusp rep irreducible at all 400 sampled points (commutant=1), mixing $\geq$0.35, reducible locus measure-zero $\Rightarrow$ case A generic, ($U_{\rm wall}$) survives; amplitude dictionary still research-level
`v39_uwall_selectors.py`	($U_{\rm wall}$) selector side closed on the explicit point: splitting type = anchor $a{=}(1,1,2)$, pardeg =0; $\det R{=}8{=}n{\cdot}a$, Spec$(Q_+){=}\{1,2,3\}{=}3\alpha{+}1$ read off the bundle; only the amplitudes c_u/c_d ($\Gamma^{\rm min}$ Hitchin) remain
`v40_harmonic_metric.py`	($U_{\rm wall}$) harmonic metric = FINITE linear algebra (polystable⇒unitary⇒$\Phi{=}0$): unique invariant form $H{>}0$, unitarised $\widetilde{M}_k$ cusp-class with $\|\widetilde{M}_k\|{\in}\{0,\frac{1}{2},\frac{1}{\sqrt{2}}\}$; the feared Hitchin PDE collapses to unitarisation + combinatorial R

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Script	What it machine-checks *(continued)*
`v41_leg_assignment.py`	(U_{wall}) final leg test: amplitudes $=\delta{=}\frac{1}{2}$ resolvent (closed); $\Lambda_\mu{>}1$ rules out holonomy moduli; harmonic holonomy $\lvert\text{diag}\rvert{=}(0,\frac{1}{2},\frac{1}{2})$ supplies $\delta{=}\frac{1}{2}$ (suggestive); honest negative on literal c_u/c_d reproduction
`v42_exterior_leg.py`	(U_{wall}) Exterior Leg Lemma: u,d share $(r,w){=}(1,1)$ so any scalar leg gives $c_u/c_d{\equiv}1$; the u/d split is the Λ^2F area $\text{Pl}_{\mathbf{1},a}(K)$, $\lVert\cdot\rVert_1{=}11$ generated $\Rightarrow \frac{5\cdot 11}{9\cdot 13}{=}\frac{55}{117}$ [I]; phys. identification [P]
`v43_exterior_bridge.py`	(U_{wall}) Λ^2F bridge: $\Lambda^2(\widetilde{M}){=}\widetilde{\widetilde{M}}$ (exterior leg $=\bar{\mathbf{3}}$) confirmed; continuous action $\neq$ integer $\text{Pl}(K)$ $\Rightarrow$ bridge is the discrete H2 invariant ($\det R{=}8$, $\text{Spec}(Q_+)$ confirmed), not continuous
`v44_carrier_exterior.py`	(U_{wall}) Lie-level grounding: $16{=}\Lambda^{\text{even}}(5){=}1{+}10{+}5$; $u^c{\in}\Lambda^2(5)$, $d^c{\in}\Lambda^4(5)$ $\Rightarrow$ exterior degrees $(2,4)$, sum$=6{=}\lvert R^+(A_3)\rvert$, diff$=2{=}\lvert\mathbb{Z}_2\rvert$; the exterior leg is the carrier's own Λ^2 grading (no special math)
`v45_family_exterior.py`	(U_{wall}) the "11" is Pascal: $\lvert\text{Pl}(K)\rvert{=}(1,4,6){=}\{\Lambda^0,\Lambda^1,\Lambda^2\}(\mu_4)$; $\lVert\text{Pl}(K)\rVert_1{=}11{=}\sum_{k\le 2}\binom{4}{k}{=}16{-}g_{\text{car}}$ (cumulative ext of μ_4 up to $\deg u^c{=}2$); $15{=}\dim\mathfrak{su}(4){=}10{+}5{=}N_{\text{fam}}g_{\text{car}}$
`v46_grand_mass_volume.py`	absolute mass-scaling is [I], not a product rule: sector dets $(\varphi_0^{\text{ret}})^6,(\varphi_0^{\text{ret}})^9,(\varphi_0^{\text{ret}})^{10}$ (exponents $\lvert R^+A_3\rvert, N_{\text{fam}}^2, \mathcal{A}_\Lambda$) $\Rightarrow \det M_{\text{SM}}{\sim}(\varphi_0^{\text{ret}})^{25}{=}(\varphi_0^{\text{ret}})^{g_{\text{car}}^2}$; Q-rows $(4,5,6)$ sum $15{=}\dim A_3$; $25{+}15{=}40{=}\sum L$
`v47_selection_theorem.py`	Boundary Carrier Selection (Thm A): $\mu_4{\to}A_3$, $N_{\text{fam}}{=}3$; 16-spinor$\to D_5$; $q(D_5){+}q(A_3){=}2$, glue $4{=}\lvert\mu_4\rvert$; only $D_5{\oplus}A_3$ meets all four conditions ($D_8, E_7{+}A_1, E_6{+}A_2, A_4{+}A_4, A_8$ excluded) [L]/[P]
`v48_em_ward.py`	EM boundary Ward (Thm C): $F_{U(1)}{=}$ Maxwell α^3 $-$ Calderón $2c_3^3\alpha^2$ $-$ transport $8b_1c_3^6\log\frac{1}{\varphi_{\text{seam}}}$; $8b_1{=}\frac{4}{5}M{=}\frac{164}{5}$, $-\frac{5}{4}{=}q(D_5)$ [I]; Ward origin [P]
`v49_readout_rigidity.py`	Readout Rigidity (Thm U2): $c_u/c_d{=}\frac{55}{117}$ is constant on the discrete selector stratum $\mathcal{S}_{8,Q}$ $\Rightarrow$ no point-uniqueness needed; $(U_{\text{wall}}){=}U_{\text{unitary}}{+}U_{\text{H2}}{+}U_{\Lambda^2}{+}U_{\text{point}}$, only U_{point} open
`v50_q_geometry.py`	Q geometry (Thm Q): $Q_+{=}$ diag$(3,2,1)$-block $\chi{=}(t{-}1)(t{-}2)(t{-}3)$, Q_- $\chi{=}t(t^2{-}3)$; Q unique under rows$(4,5,6)$/cols$(9,5,1)$/spectra; geom. origin [P]
`v51_boundary_half_step.py`	glue norms $=(g_{\text{car}},N_{\text{fam}})/\lvert\mu_4\rvert$: $q(D_5){=}\frac{5}{4}$, $q(A_3){=}\frac{3}{4}$; four ops forced $\to$ {sum$=2$ root norm, diff$=\frac{1}{2}{=}\lvert\mathbb{Z}_2\rvert/\lvert\mu_4\rvert{=}\delta$, prod$=\frac{15}{16}$, ratio$=\frac{5}{3}$}. So $\delta{=}\frac{1}{2}$ is the carrier$-$family count over the glue index, not chosen [I]
`v52_pencil_endpoints.py`	$K{+}xQ$ pencil endpoints: $P(-1){=}2{=}\lvert\mathbb{Z}_2\rvert$, $P(0){=}4{=}\lvert\mu_4\rvert$, $P(1){=}20{=}\det L$, $P(2){=}68{=}2p_5(a)$; $\det(K{-}Q){=}2$, tr $=N_{\text{fam}}$ (audit ext. of v37) [I]
`v53_compiler_core.py`	Big picture: the whole integer skeleton from $(g_{\text{car}},N_{\text{fam}}){=}(5,3)$ — rank $E_8{=}8$, $\lvert\mathbb{Z}_2\rvert{=}2$, Pythagorean $\Delta_Y{=}g_{\text{car}}^2{=}25{=}N_{\text{fam}}^2{+}\lvert\mathbb{Z}_2\rvert{\cdot}\text{rank}\,E_8{=}9{+}16{=}\dim S^+$ (diff. of squares); anchor char-poly $(t{-}1)^2(t{-}2)$ unique [I]
`v54_seam_horizon_keystones.py`	The "8" in $c_3{=}\frac{1}{8\pi}$ overdetermined and gravitationally aligned ($2\lvert\mu_4\rvert{=}\text{rank}\,E_8{=}h(D_5){=}\varphi(30){=}8\pi$ Hawking/Einstein); one boundary transport operator (spec $\{1,(2/3)^6,(1/3)^6\}$, $\lambda_2{=}(2/3)^6$) governs *both* SM flavor gap and horizon Page recovery [I]
`v55_coxeter_cycle.py`	Computed E_8 Coxeter element: order exactly $30{=}\lvert\mathbb{Z}_2\rvert N_{\text{fam}}g_{\text{car}}$, eigenphases $=$ exponents $=$ totatives of 30, so rank $E_8{=}8{=}\varphi(30){=}$ live phases of the order-30 cycle; one α^{-1} sets EM, S_{dS} and Λ ($S_{dS}\rho_\Lambda{=}32\pi^4$) [I]/[L]
`v56_unique_attractor.py`	Gapped boundary transport (gap $6\log\frac{3}{2}{>}0$) $\Rightarrow$ *unique* attracting fixed point, geometric rate $(2/3)^6$ (two starts $\to$ same direction); Coxeter in $\lvert\mu_4\rvert{=}4$ planes; entropy reset $=$ rank-1 projector [I]/[L]
`v57_horizon_crosslinks.py`	Horizon cross-links: $c_3{=}\frac{1}{8\pi}$ is the Einstein/Jacobson coefficient ($1/4{=}1/\lvert\mu_4\rvert$ entropy density); Hod QNM $\omega_R/T_H{=}\ln 3{=}\ln N_{\text{fam}}$; Hawking power denom $1920{=}\lvert W(D_5)\rvert$; one α^{-1} for EM/S_{dS}/Λ/Hubble [I] (+ [P] identification)

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Script	What it machine-checks *(continued)*
`v58_seam_horizon_chain.py`	Seam-horizon chain (precise: seam = abstract normaliser, locally realised as a horizon): one-sided S^2 Gauss–Bonnet $c_3{=}1/(2{\cdot}4\pi){=}\frac{1}{8\pi}$; KMS $4c_3\kappa$; $S_{BH}{=}M^2/2c_3$; RT in seam units. Seam–Horizon Theorem (area-law from the seam kernel) [A] open
`v59_area_law_evidence.py`	Area-law necessary condition met: a reflection-positive Gaussian (Calderón-type) kernel yields an area law (gapped block-EE saturates; gapless $\sim \frac{c}{3}\log L$), by the *same* gap that fixes the attractor; $8{=}\lvert\mathbb{Z}_2\rvert\lvert\mu_4\rvert$. The c_3 coefficient stays [A] open
`v60_lambda_metrology_branch.py`	Λ branch selection: $(8\pi)^2\delta_{\text{top}}{=}\frac{3}{4\pi^2}$, mis-scale $2c_3/\delta_{\text{top}}{=}\frac{64\pi^3}{3}{=}661.47 \Rightarrow G_N$ pinned as output; 123 orders = 119.028+3.920; ACT DR6 birefringence $0.215°{\pm}0.074°$ vs $0.2424°$ (0.37σ) [I]/[N]
`v61_cft_bridge.py`	Boundary-CFT mirror: level-1 WZW central charges $c(E_8){=}8$, $c(D_5){=}5$, $c(A_3){=}3$ = (rank $E_8, g_{\text{car}}, N_{\text{fam}}$); conformal embedding $c_{\text{coset}}{=}0$; $SU(3){\leftrightarrow}SU(4)$ reconciliation $N_{\text{fam}}{=}\lvert\mu_4\rvert{-}1$, $4{\to}3{+}1$ (one geometry $\mathbb{P}^1\backslash\mu_4$, two sides) [I]/[L]
`v62_data_scorecard.py`	TFPT vs 2024/25 data: α^{-1}, $\sin^2\theta_{12}$, β_{rad} match; $\sin^2\theta_{13}$ (2.0σ) and Starobinsky n_s vs DESI $(\sim 2\sigma)$ mild tensions; θ_{23} open; $r{\approx}0.004$ future falsifier [N]
`v63_seam_engineering_index.py`	Exact Seam-Engineering Index $\Xi{=}2\lVert V\rVert/\Delta{=}\frac{31}{24\pi^2\log(3/2)}{\approx}0.323$ $(2\lVert V\rVert{=}\frac{31}{4\pi^2}$, $248{=}\dim E_8$, $31{=}1{+}h^\vee)$, $\Delta_{\text{eff}}{\approx}1.648$; four-class possibility taxonomy [I]/[A]
`v64_causal_boundary_nogo.py`	Causal Boundary Engineering *conditional no-go*: gapped transfer has no periodic orbit (discrete no-CTC); RP+OS+gap $\Rightarrow$ ANEC $\Rightarrow$ Topological Censorship $\Rightarrow$ no traversable shortcut; only ER=EPR bridge survives [I] (core)/[P] (chain)
`v65_falsification_layer.py`	Prediction layer with explicit failure thresholds: 3 core ($v_{\text{GW}}{=}c$, no 4th gen, seam=8), 4 observational (β, r, n_s, θ_{12}), no-go + unitarity; $N_\star$ demoted to [P] input; none violated [N]
`v66_e8_casimir_degrees.py`	Organizing simplification: the compiler atoms *are* the E_8 Casimir degrees $\{2,8,12,14,18,20,24,30\}$ $(2{=}\lvert\mathbb{Z}_2\rvert$, $8{=}$rank, $14{=}\dim G_2$, $20{=}\det L$, $24{=}\lvert W(A_3)\rvert$, $30{=}h(E_8))$; $\sum{=}128{=}2^7$ (dS prefactor), $\sum \exp{=}120{=}\lvert R^+(E_8)\rvert$ [L]
`v67_area_law_coefficient.py`	Central-theorem *structural closure*: Fursaev–Solodukhin $S{=}4\pi kA$, $k{=}c_3/2 \Rightarrow$ $S{=}2\pi c_3 A{=}\frac{1}{4}A$ $(2\pi c_3{=}\frac{1}{4})$; $c_3{=}\frac{1}{8\pi}$ is the *unique* value with replica-coeff = BH $\frac{1}{4}$. Reduces to the one coeff $k{=}c_3/2$ (then *forced*, v73) [I]/[L]
`v68_seeley_dewitt_residual.py`	Residual *resolved as an anchor*: $a_2{=}{-}\frac{d}{192\pi^2}R \Rightarrow$ the *absolute* $\frac{1}{16\pi G}$ is UV-sensitive $(\propto d\Lambda^2 f_2)$, so the *scale* $1/G$ is the one dimensionful anchor (the dimensionless $k{=}c_3/2$ is forced, v73); cutoff-indep. R^2/scalaron derived [I]/[L]
`v69_d4_q_geometry.py`	D_4-equivariant Q-geometry (gate [P]$\to$[L]): on $H_1(\mathbb{P}^1\backslash\mu_4){=}B_1{\oplus}E$, $Q_+{=}3w{+}1{=}\operatorname{diag}(1,2,3)$ (cusp weights on $\mu_4{=}\mathbb{Z}_4$ eigenspaces), $Q_-{=}E$-coupling $\sqrt{N_{\text{fam}}} \Rightarrow t(t^2{-}3)$; Σ-split $=D_4{=}\mathbb{Z}_4{\rtimes}\mathbb{Z}_2$ [L]
`v70_q_integer_lift.py`	Q integer-lift sharpened: $\mathbb{Z}_4$ puncture action R unimodular (eigenvalues $\{-1,i,-i\}$); $\det Q{=}3{=}N_{\text{fam}}$, SNF $\operatorname{diag}(1,1,3) \Rightarrow$ residual = one lattice invariant $\det Q{=}N_{\text{fam}}$, not closable by D_4 alone [I]/[L]
`v71_simple_r_bridge.py`	*Simple* R-bridge (gate #3): selector stratum $\mathcal{S}_{8,Q}$ now fully derived ($\det R{=}8{=}$rank E_8 lattice + Spec$(Q_+){=}\{1,2,3\}$ from v69) $\Rightarrow$ quark *ratios* are pure integer Plücker readouts ($\frac{55}{117}$, no monodromy, v49 rigidity); residual $U_{\text{point}}{=}$ absolute norm. = an anchor [I]/[A]
`v72_q_det_from_cusp.py`	$\det Q{=}N_{\text{fam}}$ derived from the cusp class: $\det Q{=}\lvert\operatorname{coker} Q\rvert{=}$ cusp-class order = denominator of the weights $\{0,\frac{1}{3},\frac{2}{3}\}$ (same data as Spec Q_+); coker $Q{=}\mathbb{Z}/N_{\text{fam}}{=}$ triality deck group $\Rightarrow$ integer-lift residual closed [I]/[L]
`v73_k_c3_half.py`	Central coefficient $k{=}c_3/2$ *forced*: (1/2) variational $\times$ $1/(\lvert\mathbb{Z}_2\rvert 2\pi\chi(S^2))$ Gauss–Bonnet topology $(\chi(S^2){=}2)$, both cutoff-indep $\Rightarrow$ Fursaev–Solodukhin $S{=}A/4$; only the absolute 4D Newton scale stays the anchor (v68) [I]/[L]

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Script	What it machine-checks *(continued)*
`v74_compiler_micro_lemmas.py`	Spine quotient ladder ($\frac{2}{3},\frac{3}{4},\frac{4}{5},\frac{5}{4},\frac{4}{3}$ = adjacent quotients of $(2,3,4,5)$); pencil differences $(2,16,48)$=$(\lvert\mathbb{Z}_2\rvert,\dim S^+,\Omega_{\rm adm})$ sheet→1gen→3gens; anchor QF $a^\top Ka{-}\mathbf{1}^\top K\mathbf{1}$=41−25=16=$\dim S^+$ (EM budget = mass vol + one gen); solar dual anchor $L{=}R{+}61\mathbf{e}_1^\top$, $a^\top R^{-1}\mathbf{1}{=}0$ (Sherman–Morrison rank-one invariance) [I]
`v75_upoint_to_vgeo.py`	**Gate 1 complete:** $U_{\rm point}{\to}v_{\rm geo}$. Ratios closed (v49/v71) + lepton c exact (v20) + sector products = $\varphi_0^{\rm ret}$-powers (v46); (ratios, product)⇔(individuals) bijection ⇒ absolute amplitudes fixed up to *one* scale $v_{\rm geo}$ = the same anchor as $1/G$ (v68); two $[A]$ → one [I]/[A]
`v76_gmetric_reduction.py`	**Gate 2:** Tier A (IR) closed under RP+gap (Decoupling Thm: $\Delta_{\rm eff}{=}6\log\frac{3}{2}{-}\frac{31}{4\pi^2}{=}1.648{>}0$, $\lVert V\rVert{\le}248c_3^2{=}\frac{31}{8\pi^2}$); Tier B (strict TOE) holographically reduced from bulk to a finite seam-*boundary* measure (conditional: RP+tightness ⇒ closes) [I]/[P]
`v77_e8_conformal_net.py`	**G6 route:** the seam boundary measure = the $(E_8)_1$ *lattice net* (rigorous, holomorphic $c{=}8$); level-1 $c(E_8){=}\frac{248}{31}{=}8$, $c(D_5){=}5$, $c(A_3){=}3$, embedding 5+3=8 (coset $c{=}0$); imports $G6$ into RCFT/conformal-net rigor [I]/[P]
`v78_vgeo_floor.py`	$v_{\rm geo}$ **floor:** no pure number is dimensionful ⇒ $v_{\rm geo}$ irreducible by logic; all scales are ratios × one $\bar M_{\rm Pl}$; $S_{dS}\rho_\Lambda{=}32\pi^4$ pins it from one measurement; theory at the minimum (one scale + π) [I]/[A]
`v79_review_identities.py`	Four validated review identities + one firewall: hypercharge Lucas–Binet $D_n{=}(3^n{-}({-}2)^n)/5{=}1,1,7,13,55,133$ (roots = carrier charges); Inverse Anchor Thm ($\mathbf{1}^\top M^{-1}\mathbf{1}{=}1$/atom, $a^\top M^{-1}a{=}1$ for R,K,L); MacWilliams $(1,10,5)$; $6Y^2{-}Y{-}1{=}(2Y{-}1)(3Y{+}1)$. Rejects m_p/m_e, η_B near-formulas (firewall) [I]/[L]
`v80_operator_pencil_geometry.py`	Operator-pencil geometry: Anchor Singularity Thm $\det B_{K+xQ}{=}(N_{\rm fam}x{+}\lvert\mathbb{Z}_2\rvert)(N_{\rm fam}x{+}g_{\rm car}){=}(3x{+}2)(3x{+}5)$ ($\frac{2}{3}$ Koide/gap = anchor-plane singularity); buildup chain $(3,10,16,20)$; type checker $\det B_{Q,K,R,L}{=}(9,10,16,40)$; audit: differential spectrum, $F_4{\times}G_2$ shadow $248{=}52{+}14{+}26{\cdot}7$ [I]/[L] + [P]
`v81_singular_pencil_matrices.py`	Double cover $y^2{=}\det B_{K+xQ}{=}(3x{+}2)(3x{+}5)$: Koide $x{=}{-}\frac{2}{3}$ and carrier $x{=}{-}\frac{5}{3}$ are the two branch points (deck $2{=}\lvert\mathbb{Z}_2\rvert$; disc$=81{=}N_{\rm fam}^4$; separation=1 transport period). Clearing matrices $C_{2/3}{=}3K{-}2Q$ ($D_5{\oplus}A_3$ glue, $B{=}(5,7)^\top(9,11)$, $\sum{=}240$), $C_{5/3}{=}3K{-}5Q$ (neutral, $\chi{=}(\lambda{+}2)(\lambda^2{+}\lambda{+}6)$); scalaron 7= mixed anchor disc/$N_{\rm fam}$ [I]/[L]
`v82_koide_attractor_splitting.py`	Koide attractor *forced*: the branch-divisor-preserving Möbius map fixing $q{=}2,5$ is unique, multiplier $(2/3)^6{=}\lambda_2$ of the established transfer operator ⇒ $F'(2){<}1$ attractor, $F'(5){>}1$ repeller; three postulates → one [P]. Splitting trichotomy disc 81/49/40: the clean split is non-generic (anchor-first) [I] + [P]
`v83_e8net_holomorphic_uniqueness.py`	Holomorphy pins $(E_8)_1$: #primaries $=\det$ Cartan (D_8:4 vs E_8:1), and a holomorphic $c{=}8$ chiral CFT is the lattice theory of the *unique* even unimodular rank-8 lattice (Minkowski–Siegel mass $1/\lvert W(E_8)\rvert$); carrier Pascal truncation $K{=}(g{-}1)/2$ = half-spinor split [L]/[I]
`v84_frozen_registry.py`	Blind-prediction registry frozen 2026-06-09 (25 digits, machine-enforced lock formula↔value): exactly *one* θ_{12} prediction of record; r, n_s as $N_\star$ *bands*; conditional entries carry their hypothesis by name [N]
`v85_master_cover.py`	Master-cover theorem: $\det B$ is GL(2)-covariant on span$\{K,Q\}$ ⇒ exactly *one* double cover up to Möbius (disc = $N_{\rm fam}^4\det(G)^2$); the disc-81 family is its orbit ($-\frac{8}{3}{=}{-}{\rm rank}E_8/N_{\rm fam}$ = carrier − one period); μ_4 is *not* a 4:1 cover (ladder of double covers); branch trace fixes only the scalaron *scale* [I]/[L]
`v86_nstar_reheating.py`	$N_\star$ band → point [P]: $\Gamma{=}4M^3/48\pi\bar M^2{=}128$ GeV, $T_{\rm reh}{=}9.6{\times}10^9$ GeV, $N_\star(0.05/{\rm Mpc}){=}51.4$ ⇒ $n_s{=}0.9611$, $r{=}0.0045$ (inside the frozen band); honest tensions: n_s $+0.9\sigma$, A_s dichotomy (slow point -11.4σ ⇒ near-instant reheating or exponent fails) [P]

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Script	What it machine-checks *(continued)*
v87_bulk_ uniqueness_ reduction.py	Red-team Target A residual (ii) conditionally closed: holomorphic $\Rightarrow$ Rep $=$ Vect $\Rightarrow$ 2D bulk *unique* (LR/KLM/BKLR); machine contrast $SO(16)_1$ admits *six* modular invariants (incl. both E_8 pairings) $\Rightarrow$ Target A $=$ one residual [L] + [P]/[A]
v88_cp_phase_ audit.py	Target-D CP residual quantified: frozen $\delta{=}\pi/3{+}3\lambda^2$ survives at $+0.98\sigma$ vs γ_{PDG}; the $0.07°$-near alternative $\pi/3{+}2\lambda^2$ recorded as look-elsewhere trap, *not* adopted; decision at $\sigma_\gamma{\le}0.96°$ [N]
v89_carrier_index_ lemma.py	Carrier Index Lemma: KLM $\mu_A{=}[B{:}A]^2\mu_B \Rightarrow$ Jones index $[(E_8)_1{:}(D_5)_1{\times}(A_3)_1]{=}4{=}\lvert\mu_4\rvert$ (glue order *is* the index); μ-additivity $16/4^2{=}1 \Rightarrow$ holomorphy *follows*; Gate A (H)$\Leftrightarrow$(H$'$) index-4 extension [L]
v90_conical_defect_ chain.py	Fursaev–Solodukhin factor 4π *derived* (smoothed-cone Gauss–Bonnet, exactly smoothing-independent); replica $S{=}4\pi kA$, $k{=}c_3/2 \Rightarrow S{=}A/4 \Leftrightarrow c_3{=}\frac{1}{8\pi}$ (sympy-solved); residual isolated to "seam determinant $\Rightarrow$ EH form" [I] + [A]
v91_spine_ tetrahedron.py	Spine tetrahedron $\{2,3,4,5\}$ = anchor + zeroth moment; edges/faces/volume = the integer grammar, volume $120{=}\lvert R^+(E_8)\rvert$; $240{=}\lvert\mu_4\rvert\lvert E(K_4)\rvert\lvert E(K_5)\rvert$ with K_6 negative control; dual cuts typed tautological; sub-grammar incompleteness honest [I]
v92_glue_ uniqueness.py	Glue uniqueness: of all 15 subgroups of the discriminant form exactly two Lagrangian glues (the two *chiralities*, swapped by either single spinor conjugation, fixed by the simultaneous one) and exactly one halfway extension $= SO(16)_1$; tower carrier$\to SO(16)_1\to(E_8)_1$, nothing else [I]/[L]
v93_koide_ relaxation_toy.py	Koide relaxation toy (P2 narrowed): basin lemma (every physical configuration in the attractor basin); exact rate $(2/3)^6$; source at $\rho{=}{-}\varphi_0^{\mathrm{ret}}/24$ to 0.8% ([P]); honest negatives: flow time $t{=}2.84\notin\mathbb{Z} \Rightarrow$ the missing object is a *continuous* generator [I]/[N] + [P]
v94_sheet_ diamond.py	Sheet diamond: all four flavor operators on *one* surface $M(s,t){=}R{+}Q\,\mathrm{diag}(s,t,t)$ (pencil = the cut $(1{+}x,x{-}1)$; diagonal-cut disc 153 negative control); winding line $\det B{=}2\det$ for all s; cofactor seam normal $(5,-9,6)$; Plücker canonicalisation: 11 *and* 26 both live in K; spine-quotient firewall *rejected* [I]
v95_centered_ diamond.py	Centered cross: $Q{=}U{+}V$, $R/L{=}C{\mp}U$, $K/F{=}C{\mp}V$ (five objects $\to$ three); $\mathrm{Spec}\,V{=}\{0,1,2\}{=}N_{\mathrm{fam}}{\cdot}\mathrm{cusp}$; $\det C{=}14$, $\sum C{=}31{=}2^{g_{\mathrm{car}}}{-}1$; $\mathrm{Pl}_R(C){=}7(2,3,1)\parallel\mathrm{Pl}_R(L){=}10(2,3,1)$; G_2 reading audit-typed [I]
v96_branch_kernel_ selection.py	Branch-kernel selection (P1 unblocked): rank-1 anchor blocks at the branch points have *integer* kernels (carrier kernel $=\mathbf{1}$); collapse lemma $\Rightarrow(-1,1,0)$: up $=-$down, *lepton pairing* $=0$ (leptons on the ramification); sector zero ladder (up $-\frac{3}{2}$ doubly, down 0 & $-\frac{9}{5}$); anchor-placement controls [I]/[L]
v97_sheet_ conjugation_ bridge.py	Sheet-conjugation bridge (P1 $\to$ one [P]): the anchor forces the cusp conjugation uniquely $\Rightarrow$ integer T_A, $\det{=}{-}1$; $a{=}e_2{+}e_3$ (conjugation-symmetric); one deck action on $\mathbb{R}^3/\,\mathrm{span}\{\mathbf{1},a\}$; $\langle T_A,\Sigma\rangle{=}D_4$; sheet-index lemma (intertwiner dets even, min $2{=}\lvert\mathbb{Z}_2\rvert$); remaining [P]: Q_+ grading $=A_3$ discriminant grading [I]/[L] + [P]
v98_discriminant_ dictionary.py	Dictionary *derived* (P1 closed modulo GATE.QGEO): $G{=}T_A\Sigma$ acts integrally on the cusp basis ($Ge_1{=}{-}e_1$, $Ge_3{=}e_2$, $Ge_2{=}{-}e_3$ $= B_1{\oplus}E$); cusp-0 line = self-conjugate $\mathbb{Z}_4$-class-2 line; $T_AGT_A^{-1}{=}G^{-1}$ $(k{\mapsto}{-}k)$; $q_{A_3}(1){=}q_{A_3}(3){=}\frac{3}{8}$, $q_{A_3}(2){=}\frac{1}{2}$; reflection classes = glue-swap vs glue-fix parities $\Rightarrow$ P1 carries no separate [P]; residual = the one existing Q-geometry gate [I]/[L]
v99_koide_flow_ time.py	Koide flow time: canonical generator $\dot q{=}(\Delta/N_{\mathrm{fam}})(q{-}2)(q{-}5){=}(\Delta/N_{\mathrm{fam}})\det B(q)$, time-1 map $=F$ [I]; non-integrality of $t{=}2.84$ is *data-fragile* (Q crosses $\frac{2}{3}$ at $+0.43\sigma(m_\tau)$, t passes every integer ≥ 3 within $+0.5\sigma$) [N]; conditional steps: $n{=}2$ excluded (-2.9σ), $n{=}3{=}N_{\mathrm{fam}} \Rightarrow m_\tau{=}1776.9427\,\mathrm{MeV}$ $(+0.14\sigma)$, decision at $\sigma(m_\tau){\sim}0.01\,\mathrm{MeV}$ [P]; ln(46080) scale reading destroyed by controls (firewall)

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Script	What it machine-checks *(continued)*
`v101_horizon_anchor.py`	The maximal black hole is the anchor (SdS in seam units): Nariai cubic $t^3{-}3t{+}2{=}(t{-}1)^2(t{+}2)$, roots $(1,1,-2)$ = traceless anchor; $S_{\text{Nariai}}/S_{dS}{=}\frac{2}{3}{=}|\mathbb{Z}_2|/N_{\text{fam}}$ (per horizon $\frac{1}{3}$); interpolation $(x^2{+}1)/\Phi_3(x)$; three-sheet total $|\mathbb{Z}_2|S_{dS}$ conserved; $Q_{\text{geom}} \in [\frac{3}{8}, \frac{1}{2}]$ = the $SU(4)_1$ weights; mass-line cover $(1{-}3m)(1{+}3m)$, deck = horizon swap; $\tau_{BH}{=}128\, g_{\text{car}} M^3/c_3$; six atom landings, zero free parameters [I] + [P] (bulk reading)
`v102_seam_orientation.py`	One orientation: the anchor is the *stationary repeller* in both sectors — flavor relaxation = gradient flow of a cubic potential with critical points = the branch points, curvatures $V''(2){=}{+}\Delta$, $V''(5){=}{-}\Delta$ (= the gap), inflection at scalaron/2, Lyapunov rate Δ constant; SdS entropy has Nariai as its *unique* stationary point, curvature $\frac{2}{9}{=}|\mathbb{Z}_2|/N^2_{\text{fam}}$; evaporation = entropy ascent away from the anchor [I]; "one variational principle of the seam" stays a reading [P]
`v103_trisection_normal_form.py`	Trisection normal form (the canonical coordinate): SdS cubic $\Leftrightarrow \cos 3\theta{=}{-}3m$ (angle trisection; $\mathbb{Z}_3$ deck = triality); $m{=}\cos\psi/N_{\text{fam}}$ and $S_{\text{tot}}/S_{dS}{=}\frac{4}{3}{-}\frac{2}{3}\cos\frac{2\psi}{3}$ — *one* cosine of glue atoms; canonical curvature $(\frac{2}{3})^3{=}(|\mathbb{Z}_2|/N_{\text{fam}})^{N_{\text{fam}}}$ (Koide constant to the family power); invariant slope $d\sigma/dm|_N{=}{-}\frac{8}{9}{=}{-}\operatorname{rank} E_8/N^2_{\text{fam}}$; bridge: flavor rate $(2/3)^{2N_{\text{fam}}}$ vs gravity curvature $(2/3)^{N_{\text{fam}}}$, missing = one *clock* [I] + [P]
`v104_nariai_clock.py`	The *classical* Nariai clock is the anchor (third appearance): static pin $\phi{=}\rho$ solves the dS_2 static-patch equation with $m^2{=}{-}2\Lambda{=}{-}|\mathbb{Z}_2|\Lambda$ exactly (Ginsparg–Perry tower: one negative mode); in Hubble units $\chi_{\text{clock}}{=}(\lambda{-}1)(\lambda{+}2)$ = the anchor quadratic, eigenvalues $\{1,-2\}$; Nariai cubic = $(t{-}1)\cdot\chi_{\text{clock}}$; entropy-deviation rate $2H{=}|\mathbb{Z}_2|H$; honest (2/3)-test *negative* for the classical clock — the quantum clock is the one remaining [P] of the seam variational principle [I] + [P]
`v105_residual_inventory.py`	The residual inventory: *one constant* ($2/3{=}|\mathbb{Z}_2|/N_{\text{fam}}$ exact in seven places across both sectors), *one anchor* (three dynamical appearances: configuration, Nariai roots, clock spectrum), *relocation theorem* (cosmological quantum clock suppressed by the Λ hierarchy, 121.5 orders $\approx 2\alpha^{-1}/\ln 10 \Rightarrow$ the clock must be the seam transfer operator — one [P] identification), and the *complete machine-pinned gap list*: exactly five structural objects + one scale + π; a sixth gap would fail the script [I] + [P]
`v106_review_validation.py`	External-review validation: seed normal form $\varphi_0^{\text{ret}}{=}\frac{|\mu_4|}{N_{\text{fam}}}c_3{+}\Omega_{\text{adm}}c_3^{|\mu_4|}$ exact [I]; hypercharge moment $\operatorname{Tr} X^2{=}120{=}5!$ computed from the charge table; factorial spine $5!{=}|R^+(E_8)|{=}\sum$exponents, $240{=}2{\cdot}120$, $1920{=}2^4{\cdot}5!$; degree-2 inventory (Pascal $K{=}2$ unique at $g{=}5$, glue norms sum 2, $c_3{=}\frac{1}{2}{\times}\frac{1}{4\pi}$, $\binom{5}{2}{=}10$); named [P] hypothesis *Quadratic Boundary Locality* ($\Rightarrow K{=}2$ forced; would upgrade the Pascal selection to [L]); audit $240{=}16{\times}15$ non-unique [I] + [P]
`v107_quantum_clock_target.py`	Quantum-clock target made quantitative: the classical Ginsparg–Perry clock decays on $\{0,-1,-2\} = -\operatorname{Spec}(V) = -N_{\text{fam}}{\times}$cusp (the transfer operator's own grading) [I]; exact target identity $(1/3)^6 = \big((2/3)^6\big)^{\log_{3/2} 3}$ — required bend $\log_{9/4} 3 = 1.355$ per weight step; coupling landscape $\kappa = c/(24\pi)\cdot\Lambda/\bar{M}^2_{\text{Pl}}$: 10^{-122} cosmological vs $1/(3\pi)$ at seam curvature (the only viable, borderline-nonperturbative regime) [I]; identification stays [P]
`v108_pascal_ladder.py`	Pascal Ladder Theorem: $2^{g-1}{=}\sum_{k\le K}\binom{g}{k} \Leftrightarrow g{=}2K{+}1$ (odd-g midpoint identity + even-g straddle) $\Rightarrow$ carrier selection $g{=}5$ *exactly equivalent* to degree bound $K{=}2$ — the Target-B residual = the one QBL hypothesis; neighbour worlds: $K{=}1$ one-family, $K{=}3$ nine-family, $K{=}4$ inconsistent; only $K{=}2$ also satisfies $g{+}N_{\text{fam}}{=}8$ (v14 overdetermination) [I]/[L] + [P]
`v109_sheet_pairing.py`	Sheet-Pairing Lemma (QBL anchored): character-exact multisets from $(\pm\frac{1}{2})^5$ — no scalar within a sheet (zero-weight mult of $S^+{\otimes}S^+ = 0$; odd g flips parity); $S^+{\otimes}S^-{=}\Lambda^0{\oplus}\Lambda^2{\oplus}\Lambda^4$ exactly ($256{=}1{+}45{+}210$, zero-modes $(1,5,10)$ = the code); sheet-diagonal = odd forms ($2\Lambda^1{+}2\Lambda^3{+}\Lambda^5$); tower tops at Λ^{2K}, $K{=}2 \Rightarrow$ the seam scalar kernel is necessarily a *sheet pairing* [I] + [P]

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Script	What it machine-checks *(continued)*
`v110_calderon_sheet.py`	Calderón-sheet selection theorem: a Calderón involution ε on $S^+ \oplus S^-$ certifies a scalar two-point datum *iff* it is sheet-odd; sheet-odd $\Rightarrow$ exactly $2{=}\lvert\mathbb{Z}_2\rvert$ invariant kernels = the two Lagrangian glues; ladder genericity ($g{=}3,5,7$): sheet-oddness forces $K{=}(g{-}1)/2$, *not* $g{=}5$ (anti-overclaim); QBL residue split into *interface* ("ε sheet-odd", natural from the one-sided collar) + *analytic core* (slot-degrees ≤ 2) [I] + [P]
`v111_quadratic_transport.py`	Quadratic-Transport Theorem: in the integer Jordan–Wigner model linears are sheet-odd, the 45 quadratics ($=\mathfrak{so}(10)$, the Λ^2 term of 1+45+210) sheet-even, generate the code from the vacuum *and* span the full $\mathrm{End}(S^+){=}256$ at word length 2 $\Rightarrow$ *transport degree exactly 2* (degree ≤ 1 generates nothing) — the transport half of the QBL core is a theorem; $g{=}3$ control: degree selected, not rank [I] + [P]
`v112_selfcounting_channel.py`	Self-Counting Channel: $w \mapsto (w, -w)$ is a bijection code states $\leftrightarrow$ neutral certified kernels, so #kernels $= 2^{g-1}$ *exactly*; graded by pair-degree the same set has sizes $\binom{g}{m}$ $\Rightarrow$ the Pascal closure $2^{g-1}{=}\sum_{m\leq K}\binom{g}{m}$ is *two countings of one set* (an identity, not a requirement); Hodge fold $\binom{g}{2j}{=}\binom{g}{\min(2j,g-2j)}$; QBL residue = one input ("a single scalar 2-point kernel"), selection carried by rank-8+integrality [I] + [P]
`v113_quasifree_kernel.py`	One kernel is the whole net: carrier $(D_5)_1{\times}(A_3)_1 = SO(10)_1{\times}SO(6)_1 = 16$ free Majoranas, $c{=}5{+}3{=}8$ at every tower level (only the glue grows, index $2{\times}2{=}4{=}\lvert\mu_4\rvert$); Wick/Pfaffian exact (all 210+210 higher correlators = Pfaffians of the one kernel); kernel = Calderón involution, rank $P{=}5{=}g_{\mathrm{car}}$, seam level $8{=}\,\mathrm{rank}\,E_8$ (c = rank of the kernel); vacuum unique $\Rightarrow$ QBL input = R2 premise: gate closes $\Rightarrow$ carrier choice closes [L] [I] + [P]
`v114_torsion_delta.py`	Torsion normal form + $\delta{=}\frac{1}{2}$ theorem: flatness of the (U_{wall}) $\mathbb{Z}_4$-family $\Leftrightarrow$ $(MU)^4{=}\mathbf{1}$ (μ_4 = order of the twisted cusp generator); on the involutive branch ($T^2{=}\mathbf{1}$) the cusp trace forces diag $M{=}(0, \frac{i}{2}, -\frac{i}{2})$ with spec $\{1, \omega, \omega^2\}$ automatic $\Rightarrow$ $\delta{=}\frac{1}{2}$ exact and constant (v20's hand value = torsion identity); census: $\{1, i, -i\}$ branch varies, v33 point in its $d_1{=}0$ slice; residue = bundle-side branch selection [I] + [N] + [P]
`v115_anchor_residue.py`	The anchor pins the residue: μ_4-average = diagonal projection $\Rightarrow$ anchor splitting $\Leftrightarrow$ diag $A_0{=}a/\lvert\mu_4\rvert$; off-weights uniquely $(8,0,5)/144$ $({=}\,(\mathrm{rank}\,E_8, 0, g_{\mathrm{car}})/(\lvert\mu_4\rvert N_{\mathrm{fam}})^2$, total $13{=}\Delta_Q$); *exact* residue $A_0^\star$ with $\chi{=}\lambda(\lambda{-}\frac{1}{3})(\lambda{-}\frac{2}{3})$ is the RH solution ($\lVert M_\infty{-}\mathbf{1}\rVert{\sim}10^{-10}$); anchor forces $d_1{=}0$ $\Rightarrow$ harmonic diagonal anchor+torsion-forced [I] + [N]
`v116_resonance_uniqueness.py`	Resonance theorem (v115 [N]$\to$[I], linear — no Gröbner): twisted μ_4 averages collapse to $B_1{=}4a_{12}E_{12}{+}4\overline{a_{13}}E_{31}$; anchor exponents diag$(2,1,1)$ give resonance gap 1 with obstruction $4\overline{a_{13}}$ $\Rightarrow$ $M_\infty{=}\mathbf{1} \Leftrightarrow a_{13}{=}0$; + $(8,0,5)/144$ + Diagonaleichung $\Rightarrow$ flat anchor locus = *one* gauge orbit = exact $A_0^\star$ — U_{wall} bundle datum algebraically unique; sharpness control $\lVert M_\infty{-}\mathbf{1}\rVert \sim 8.5\lvert a_{13}\rvert$ [I] + [N]
`v117_monodromy_weyl_a3.py`	The monodromy is $W(A_3)$: the unitarised holonomy of $A_0^\star$ is *exact* (entries in $\frac{1}{2}\mathbb{Z}[i]$), diag $\widetilde{M}_0{=}(0, -\frac{i}{2}, \frac{i}{2})$ $\Rightarrow$ $d_1{=}0$, $\delta{=}\frac{1}{2}$ are matrix entries; $\langle U, \widetilde{M}_0\rangle$ enumerated exactly: order 24, statistics $(1,9,8,6)$, characters $(3,-1,0,1) = S_4{=}W(A_3)$ (standard$\otimes$sign) — the flavor wall realises the A_3 glue factor as its Weyl group; ODE match $3.5{\times}10^{-10}$ [I] + [N]
`v118_hexagon_family_dictionary.py`	First exact H2 piece: $\mathrm{spec}(\widetilde{M}_0) \cup \mathrm{spec}(-\widetilde{M}_0) = \mu_6$ (hexagon = family spectrum + sheet twist, $\mathbb{Z}_6{=}\mathbb{Z}_2{\times}\mathbb{Z}_3$); $\det(\mathbf{1}{-}t\widetilde{M}_0){=}1{-}t^3$ $\Rightarrow$ $\det_\mp = (\frac{7}{8}, \frac{9}{8})$; lepton coefficients = resolvent determinants ($c_e{=}\lvert\mu_4\rvert/(2\det_-)$, $c_\mu{=}N_{\mathrm{fam}}/(2\det_+)$, $c_\tau{=}\lvert\mu_4\rvert\det_-/N_{\mathrm{fam}}$, product $= \lvert\mu_4\rvert/\det_+ = \frac{32}{9}$); $\langle U, \widetilde{M}_0, -\mathbf{1}\rangle$ has order $48{=}\Omega_{\mathrm{adm}}$; the address table stays open [I] + [P]

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Script	What it machine-checks *(continued)*
`v119_review_validation_2.py`	Second review validation: anchor ratio triad $(e_3,e_1,e_2)/p_0{=}(\frac{2}{3},\frac{4}{3},\frac{5}{3})$ = (Koide branch, seed gain, carrier branch; flow critical points $(2,5){=}(e_3,e_2)$); the 121 audit lemma $(\mathbf{1}{+}a)^T R(\mathbf{1}{+}a){=}121{=}11^2{=}\|\,\mathrm{Pl}(K)\|_1^2$ (orderings $\{105,121,135\}$, only canonical $=121$; audit, not load-bearing); micro-identities $\Omega_{\mathrm{adm}}{=}2p_1p_2$, $10b_1{=}1{+}p_1p_3$, $\Delta_Y{=}e_2^2{=}p_0^2{+}2\,\mathrm{rank}$; flavor surface = v94/v95 (presentational) [I]
`v120_address_table.py`	Address table (frozen v18/v20 integers): lepton words $(8,5,3){=}(\mathrm{rank}\,E_8,g_{\mathrm{car}},N_{\mathrm{fam}})$; addresses = divmod by the hexagon ($p_2{=}6$): $e{\to}(2,1)$, $\mu{\to}(5,0)$, $\tau{\to}(3,0)$, $\sum r{=}p_3$; sheet parity: τ sits at the real twisted eigenvalue -1 (= the product-rule site); quark sums: up $=p_3$, down $=p_1{+}p_3$, all quarks $=24{=}\lvert W(A_3)\rvert$, all nine $=40{=}p_1p_3{=}a^TRa$; top at the vacuum site $(0,0)$; the assignment derivation stays open [I] + [P]
`v121_address_pinning.py`	Address pinning: $L{=}R{+}2U$ (v95) $\Rightarrow$ the word table = the residue operator R + the generation-1 winding, R column 1 = the anchor a; margins = atoms (rows (e_1,rank,p_3), columns $(e_1,\Delta_Q,g_{\mathrm{car}})$); *census theorem*: among all hexagon matrices with atom margins (17 of them) exactly *one* has $\det{=}8$ — and it is R (also unique with SNF $(1,1,8)$; all 17 determinants distinct) $\Rightarrow$ the address table carries *zero* residual information; the H2 residue = the six atom margins [I] + [P]
`v122_margin_theorem.py`	Margin theorem: the *established* selectors — the D_4 annihilator $n{=}(5,-9,6)$ with $n^TR{=}(8,0,0)$ (v94), $\det R{=}8$, $\det K{=}4$ (the v71 quartet) — pin R *uniquely* among hexagon matrices ($64\to12\to1$) $\Rightarrow$ the atom margins, the anchor column and all sum rules are *consequences* (their input status is revoked); bonus: $\det L{=}20$ holds automatically on the locus (the quartet is redundant); H2 introduces *no* new residual class [I] + [P]
`v123_inventory_update.py`	Residual inventory updated (post v110–v122): thirteen new ledger rows machine-pinned; R2 carries *three* loads (metric + carrier + QBL: one theorem, three doors); the table re-pins to *five* classes — R1 clock, R2 index-4 net, R3 EH, R4′ selector establishment (replacing the *discharged* H2 dictionary), R5 Q realisation — + v_{geo}, π; a sixth class fails the script [I]
`v124_resummed_clock.py`	Resummed clock (the R1 bend in closed form): transfer eigenvalues = exactly $(1{-}n/N_{\mathrm{fam}})^{p_2}$ $\Rightarrow$ $\mathrm{rate}(n){=}{-}p_2\ln(1{-}n/N_{\mathrm{fam}})$ reproduces $\{0,\Delta,6\ln3\}$, the bend $\log_{3/2}3$ in one line; the linearisation carries slope $\lvert\mathbb{Z}_2\rvert{=}2$ relative to the classical GP clock $\{0,-1,-2\}$; the pole at $n{=}N_{\mathrm{fam}}$ forces the three-level spectrum; the R1 target is now closed-form [I] + [P]
`v125_glue_qsystem.py`	Glue Q-system: $\mathbb{C}[\mathbb{Z}_4]$ satisfies all Longo Q-system axioms exactly (associativity, unit, Frobenius, specialness $mm^*{=}\lvert\mathbb{Z}_4\rvert\mathbf{1}$) $\Rightarrow$ Jones index $4{=}\lvert\mu_4\rvert$ (the v89 value); locality = isotropy ($q(k(1,1)){=}k^2\in\mathbb{Z}$, v92); halfway sub-Q-system $\mathbb{C}[\mathbb{Z}_2]$ = the $SO(16)_1$ step, $4{=}2{\times}2$; R2 sharpens from "find" to "identify" [I] + [P]
`v126_clock_wall_bridge.py`	Clock and wall share one spectrum: the resummation weights $n/N_{\mathrm{fam}}{=}\{0,\frac{1}{3},\frac{2}{3}\}$ = exactly spec $A_0^\star$ (the parabolic MS weights); $\lambda_n{=}(1{-}\alpha_n)^{p_2}$; checkpoint 1 passed *classically* ($p_2/N_{\mathrm{fam}}{=}2{=}\lvert\mathbb{Z}_2\rvert$ = the v104 entropy rate $2H$); $\det(\mathbf{1}{-}A_0^\star){=}\frac{2}{9}$ = the entropy curvature (v102); the quantum content of R1 is isolated in the geometric tail [I] + [P]
`v127_ring_resummation.py`	The clock is a log determinant: rate $={-}p_2\,\mathrm{Tr}\ln(\mathbf{1}{-}\alpha P)$ — six identical ring (RPA) towers, *one per hexagon site*, coupling = the parabolic weight; the $k{=}1$ ring = the classical term, the $k{=}2$ ring = the first quantum correction; $\kappa_{\mathrm{seam}}{=}\frac{1}{3\pi}{=}\alpha_1/\pi$ (audit); the wall = the RPA instability at $\alpha{\to}1$; v107's "O(1) resummation" is thereby *typed* (ring/log-det), its derivation open [I] + [P]
`v128_graded_hull.py`	Graded hull + zero-mode multiplicity: the 240 E_8 roots explicit in $D_5{\oplus}A_3$+glue coordinates, coset decomposition $240{=}52{+}64{+}60{+}64$; the grading is exactly additive on *all* 6720 root-pair sums $\Rightarrow$ E_8 is a $\mathbb{Z}_4$-*graded* Lie algebra over the carrier (grading = glue = Q-system); ring multiplicity $6{=}2(2l{+}1)\vert_{l=1}$ = the sheet-doubled GP zero modes [I] + [P]

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Script	What it machine-checks *(continued)*
`v129_entropy_power_law.py`	The clock is an entropy power law: $\lambda_n{=}(S_n/S_{dS})^{p_2}$ with the three Nariai entropy levels $S/S_{dS}{\in}\{1,\frac{2}{3},\frac{1}{3}\}$ (v101); rate $= p_2 \ln(S_{dS}/S)$; triple identity $\alpha{=}n/N_{\rm fam}{=}$ MS weight = entropy-deficit share ⇒ the RPA coupling is nowhere free; orientation consistent (attractor = max entropy); R1 thermodynamic: $\Gamma{\propto}(S/S_{dS})^{p_2}$ (a Gibbons–Hawking power law, $h{=}N_{\rm fam}$ audit) [I] + [P]
`v130_born_square.py`	Born square: amplitude $\sim (S/S_{dS})^{n_{\rm zero}/2}{=}(S/S_{dS})^3$ (one $S^{1/2}$ per zero mode), probability $= \lvert A\rvert^2{=}(S/S_{dS})^6 \Rightarrow h{=}n_{\rm zero}/2{=}3{=}N_{\rm fam}$, $p_2{=}2h$ (the Born rule) — the v129 weight audit is *derived*; the 3 = the dimension of the $l{=}1$ space = the number of proper conformal Killing vectors of S^2; the $\times 2$ = the horizon pair (the v101 deck); the R1 residue = the standard zero-mode measure [I] + [P]
`v131_measure_is_area.py`	The measure is the area: $\lVert Y_{1m}\rVert^2{=}r^2{=}A/(4\pi)$ *exactly* (all three $l{=}1$ modes integrated) ⇒ the per-mode Jacobian = the mode norm $\sim S^{1/2}$ ⇒ amplitude S^3, Born S^6 — every exponent factor with a derivation origin; the 4π = the same Gauss–Bonnet primitive as in c_3, cancelling in all ratios; the R1 residue = *one* det-ratio cancellation $\det'_n / \det'_{dS} = 1{+}O(\text{deficit})$ [I] + [P]
`v132_detratio_anomaly.py`	Det-ratio anomaly = the Koide constant: $\zeta(0)\rvert_{\det'}$ of the GP sphere operator $-\Delta{-}2$ (the 3 zero modes removed) $= a_1{-}3 = \frac{7}{3}{-}3 = -\frac{2}{3} = -\lvert\mathbb{Z}_2\rvert/N_{\rm fam}$ exactly ($a_1{=}\frac{1}{N_{\rm fam}}{+}\lvert\mathbb{Z}_2\rvert$; numerical continuation $\sim 10^{-7}$) — the eighth 2/3 appearance, the first as a *spectral* anomaly; consequence $\det'(cL){=}c^{\zeta(0)}\det'(L)$; the R1 residue = *one* $\zeta(0)$ budget statement (the 4d sum) [I] + [P]
`v133_zeta_budget.py`	The ζ budget, both routes exact (Euclidean Nariai = $S^2{\times}S^2$): the reduced route gives $-\frac{2}{3}$ per sector, total $-\frac{4}{3}{=}{-}e_1/p_0$ (= minus the seed gain — two of the three triad values); the naive 4d route gives $a_2{=}\frac{161}{45}$, $\zeta(0){=}{-}\frac{109}{45}$ (*no* atom; continuation $\sim 10^{-8}$); zero modes 3+3 = the horizon pair ⇒ the budget structurally favours the *reduced* seam reading; remainder = graviton/ghost shares (Volkov–Wipf class) [I] + [P]
`v134_dual_anchor.py`	Dual anchor lemma: $d{:=}a^\top R^{-1}{=}a^\top L^{-1}{=}(-\frac{1}{2},-\frac{1}{2},1)$, $d{\cdot}\mathbf{1}{=}0$, $d{\cdot}a{=}1$, $(1,1,-2){=}{-}2d$ — the inverse flavor response *is* the Nariai root (v101); the invariance is exact via Sherman–Morrison (⇔ orthogonality to $R^{-1}\mathbf{1}{=}\frac{1}{4}(1,1,-1)$; controls: $\mathbf{1}, e_1, n$ do *not* lie on the plane); (d,n) = the dual normal pair of the flavor boundary; a third purely *algebraic* leg of the flavor↔horizon bridge [I] + [P]
`v135_det_surface.py`	Determinant surface: $\det M(s,t){=}3s(t{+}1)(t{+}2){+}t^2{+}5t{+}8$ — two s-independent iso-volume walls $t{=}{-}2{\Rightarrow}2{=}\lvert\mathbb{Z}_2\rvert$, $t{=}{-}1{\Rightarrow}4{=}\lvert\mu_4\rvert$ ($\det K{=}4$ is a *layer*); winding line $6s{+}8 \Rightarrow (8,14,20)$, $\det F{=}32{=}2^{g_{\rm car}}$; $\det B(s,0){=}2\det M(s,0)$ exactly, B-wall only at $t{=}{-}2$; row-sum axes $C{=}(7,11,13)$, $U{=}(3,3,3)$, $V{=}(1,2,3){=}\mathrm{Spec}(Q_+)$; micro-identities $e_1^2{+}e_2^2{=}41{=}10b_1$, $p_1^2{+}p_2^2{=}52{=}$ the carrier coset root count (v128) [I] + [P]
`v136_dual_normal_selector.py`	Dual-normal selector: (d,n) pins R *columnwise* — $d{\cdot}c_j{=}a_j$, $n{\cdot}c_j{=}8\delta_{1j}$, kernel = primitive $(6,8,7)$, census: each column unique (1 of 216) ⇒ R4′ shrinks from 6^9 to three column problems; the $A_0^\star$ zero mode carries $\sigma{=}(2,-9,5){=}(\lvert\mathbb{Z}_2\rvert,-N_{\rm fam}^2,g_{\rm car})$ ($\lVert v\rVert^2{=}\frac{16}{9}$, $16{=}\dim S^+$; $\mathrm{adj}{=}\frac{2}{9}P_v$ = the SdS curvature); $n{\cdot}\sigma{=}121{=}11^2$; $W(A_3)$ orbit control: the naive lift $\sigma{\to}n$ fails — a genuine mechanism, not a group element [I] + [P]
`v137_qplus_cohomology.py`	The Q_+ grading is cohomology: $H^1(\mathbb{P}\setminus\mu_4){=}\chi_1{\oplus}\chi_2{\oplus}\chi_3$ ($\omega_k{=}z^{k-1}dz/(z^4{-}1)$, character i^k, residues $\zeta^k/4$); degrees $(1,2,3){=}\mathrm{Spec}(Q_+){=}$ the A_3 exponents ($h{=}4{=}\lvert\mu_4\rvert$); cusp weights $\{0,\frac{1}{3},\frac{2}{3}\}{=}\mathrm{spec}(A_0^\star) \Rightarrow$ MS weights = cusp classes = characters = Q_+: *one* spectrum, four readouts; the QGEO residue = the naturality of the canonical map [I] + [P]

continued on next page

Script	What it machine-checks *(continued)*
`v138_vw_firewall.py`	Volkov–Wipf firewall (R1 typed closed): external 4d value $\alpha_{\mathrm{PI}}{=}-\frac{98}{45}{=}\frac{22}{45}-\frac{8}{3}$ (arXiv:2506.02142 Tab. 4.1 = VW hep-th/0003081); **edge match**: $\alpha_{\mathrm{edge}}{=}-\frac{8}{3}{=}2{\times}(-\frac{4}{3})$ — two identical edge copies (the horizon pair), per copy exactly the reduced seam budget (v133, the seed gain); bulk $\frac{22}{45}$ = the graviton gas (not the clock); difference $-\frac{38}{45}$ no atom ⇒ typing: the seam clock = an edge object, the full 4d determinant = the gravity firewall [I] + [P]
`v139_selector_triangle.py`	Selector triangle: $d{=}\frac{3}{2}a{-}2\cdot\mathbf{1}$ exactly (pure anchor data, no R) ⇒ the first selector is *derived*; the frame $(\mathbf{1}, a, \sigma)$ has $\det = 11 = \|\operatorname{Pl}(K)\|_1$, and n is the *unique* covector with atom pairings $(2, 8, 121){=}(\lvert\mathbb{Z}_2\rvert, \operatorname{rank} E_8, 11^2)$; on the constrained line the pairing is $11(11{+}t)$, a square only at the true n; review lift exact (audit): $n{=}\operatorname{reverse}(\sigma){+}\lvert\mu_4\rvert e_3$; R4′ residue = three pairings [I] + [P]
`v140_canonical_map.py`	The canonical map exists and is rigid: the plain deck U (characters $(0, 1, 3)$) cannot match H^1; the *sheet-twisted* $-U$ $((2, 3, 1))$ and the cusp exponential i^{Q_+} $((1, 2, 3))$ both carry the H^1 character set $\{1, 2, 3\}$ (the v118 sign twist); multiplicity-freeness ⇒ Schur-rigid equivariant isomorphism (unique up to scalars); i^{Q_+} pulls the Euler degree back to Q_+, $-U$ to $Q_+ \circ \rho$ — the QGEO residue collapses to *one* $\mathbb{Z}_3$ choice [I] + [P]
`v100_numerology_null_mc.py`	Numerology null test (look-elsewhere corrected): exact census of the full declared formula grammar over the 13 scored frozen-registry observables (conservative data windows) gives joint formula-fishing probability $\prod p_i = 10^{-25.8}$, robust to budget/window variation; $2{\times}2{\cdot}10^5$ Monte-Carlo pseudo-theories reach at most 5/13 hits vs. TFPT 13/13; power controls (seed perturbation, data shuffle) collapse the scorecard; all 94 500 $F_{U(1)}$ variants solved, exactly *one* hits CODATA ⇒ total $\leq 10^{-30.7}$, *conditional on the declared grammar* [N]

How to run. From the repository root, `python verification/run_all.py` (dependencies `mpmath`, `numpy`, `sympy`); each module also runs standalone, e.g. `python verification/v1_e8_glue.py`. The runner exits with code 0 iff all checks pass; see `verification/README.md` for the full claim↔script map and `verification/status_ledger.csv` for the **single status ledger** (claim ID, status, location, dependencies, script, external datum — the source of truth all six documents are written against). The [F] (Lean-formalised) carrier material lives in the canonical shipped folder `lean4-carrier-rigidity/` (at the package root; the source repository keeps the same project under `experiments/`).